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The Disabled Patient Part 1: Assistive Devices and Technology

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Introduction

Assistive technology is an umbrella term that includes assistive, adaptive, and rehabilitative devices for individuals with disabilities and also includes the process used in selecting, locating, and using them. Assistive technology (for this chapter called devices) can be defined as “any item, piece of equipment, or product system, whether acquired commercially off the shelf, modified, or customized, that is used to increase, maintain, or improve functional capabilities of individuals with disabilities” (Nicolson *et al.*, 2012). The implications for veterinary rehabilitation are as follows:

- Assistive devices can decrease effort and energy expenditure for family caregivers of animals with disabilities.
- Assistive devices can increase animal independence, and therefore decrease family caregiver burden.
- Rehabilitation veterinarians and their technicians/nurses need to educate and guide family caregivers on the use of assistive devices for their pet in order to improve the chances of correct use along with patient acceptance.

In the past 20 years we have had a surge of understanding and innovation for veterinary pain management and rehabilitation. This

chapter will endeavor to highlight some of the useful devices, including orthoses and prostheses.

History

Canine “wheelchairs” have been on the US market for around 55 years. Dr. Lincoln Parkes designed carts in 1961 for his surgical patients, usually small-breed dogs but some German Shepherds. By the late 1970s, he had formed a company—K9 Carts (Glenn Parkes, personal communication). Carts are usually custom-made using the specific measurements of the canine, feline (or other) patient. In 2008, a mass market dog wheelchair was produced (Walkin’ Wheels, <http://walkinwheels.com/>). This device fit a range of dog shapes and sizes and made dog wheelchairs easier for veterinarians to stock ready for future use (Bluebird Care, 2015).

The profession of bracing predates surgery. Bone setters and appliance makers were skilled artisans (Mich, 2014). Orthopedic surgery development caused bracing to become secondary to surgery. The correct term for braces is orthoses. Orthoses are defined as any medical device attached to the body to support, align, position, prevent, or correct deformity, to assist weak muscles, or to improve function (Mich, 2014).

Originally, blacksmiths (farriers) and armor makers were prosthetists. As the US Civil War dragged on, the number of amputations rose astronomically, forcing Americans to enter the field of prosthetics. James Hanger, one of the first amputees of the Civil War, developed what he later patented as the “Hanger Limb” from whittled barrel staves (Norton, 2007). The American Orthotic and Prosthetic Association was founded following World War I. Today’s devices are much lighter, made of plastic, aluminum, and composite materials to provide amputees with the most functional devices.

When we optimize movement and mobility for our patients, we can significantly affect their physical and mental health. Organizations such as the American Association of Rehabilitation Veterinarians, the American College of Veterinary Sports Medicine and Rehabilitation, and the recently formed Academy of Physical Rehabilitation Veterinary Technicians are educating veterinarians on the use of assistive devices, orthoses, and prostheses to improve the mobility and functionality of the veterinary patient.

Which Patients Need Assistive Devices?

Patients that have a physical injury, who are postoperative, are orthopedic or neurological, or who have other physical rehabilitation needs due to chronic conditions may be in need of an assistive device. It is necessary to assist postoperative patients in getting into and out of cages, and to help them up from a recumbent position, as well as assisting while ambulating to prevent slips, excess weight bearing on a new repair, or further pain and injury. The patient must be allowed to become accustomed to any assistive device and the procedures involved in placement and assisting ambulation (Drum *et al.*, 2014).

Patients with spinal cord injuries may have reduced body sensation and awareness (proprioception) but be ambulatory, or may be non-ambulatory paretic or fully paralyzed. It is important for the therapist to recognize the need for assisted standing even in a paralyzed patient. The cardiovascular system and the respiratory system benefit from being upright, in addition to the beneficial effects on muscle circulation (Olby, 2010). In patients who have some motor function, assisting motion is necessary in order to prevent falls as well as to provide the benefits of gait retraining and strengthening (McDonald and Sadowsky, 2002).

Figure 10.1 Sling for geriatric dog.



Types of Assistive Devices

Devices Used for Assisted Standing

Standing exercises focus on strengthening shoulder, elbow, hip, and stifle extensors and re-educating muscles needed for balance and proprioception. They can also aid lung and cardiovascular function. Several approaches can be used to facilitate standing. Particularly when a patient is large, assistive devices such as slings, physio rolls, mobility carts, wheelchairs, or lifts may be used to maintain a patient in a standing position (Drum *et al.*, 2014). Smaller patients can be supported manually to maintain a standing position (Drum, 2010).

Devices that can be used include:

- Slings (Figure 10.1)
- Therapy balls (Figure 10.2)
- Hoyer lift (Figure 10.3).



Figure 10.2 Therapy ball assisting standing.

Devices Used for Assisted Ambulation

Devices for assisted ambulation can be used to help weak or paretic patients. The devices used to assist companion animals may support or protect weak, unstable joints or feet

(orthoses, boots); replace a missing portion of a limb (prostheses); or facilitate locomotion (slings, carts, ramps, steps) (Marcellin-Little and Levine, 2014). Assistive devices may also decrease the complications present in recumbent patients, including decubitus ulcers (Adamson *et al.*, 2005).



Figure 10.3 Dr. Tomlinson with a patient in a Hoyer lift.

Pets wearing assistive devices should not be left unsupervised. Several sessions may be needed to establish acclimation. Low-stress handling techniques and positive reinforcement should be observed (see Chapter 9). In general, hair removal should be avoided as hair provides skin protection, but some devices may require that hair be clipped. Devices should be inspected for wear and damage and cleaned daily. The skin should be inspected twice daily. Cleaning is especially important if there are any incisions, wounds, or open sores in contact with the assistive device.

Boots

Boots can be used to protect paws from environmental hazards, for example to protect working animals from ice and snow, rocky terrain, or hot surfaces. Patients with proprioceptive deficits may wear boots to provide more traction, protect skin, or to stop them wearing their nails down to the quick (Prydie and Hewitt, 2015). Wearing boots at home can provide traction on slippery floors and breathable boots can be left on during

the day. Boots act as sock-like coverings and are securely fastened by Velcro straps at the top. Most have a rubber sole to prevent slipping and are machine washable. The boots should be removed several times daily to assess the skin condition, especially in neurologically impaired patients, and, if possible, when performing therapeutic exercise to increase weight bearing and proprioception through the bottom of the pads (Adamson *et al.*, 2005). A proper fit is essential, and appropriate education instructions for skin care and rehabilitative exercises must be communicated to the owner.

Customized grip socks are also available. These have rubber tread under the paw surface or over the whole foot, including dorsally. Old socks may also be used to help provide padding; however, if the top is secured with tape care should be taken to avoid cutting off circulation (Marcellin-Little and Levine, 2014).

One study that evaluated a custom orthotic boot found that the boot was effective at immobilizing the distal extremity and reducing contact between a wound and the boot, which allowed for complete healing by contraction and epithelialization (Hardie and Lewallen, 2013).

Where Do I Get Boots?

- **Pawz™** (<http://pawzdogboots.com/boots/>) are a thin rubber boot without padding so the dog is able to feel the ground. Tip: The animal's feet can get hot so use foot powder to prevent moisture build-up or put a few small holes in the boot with a pin to create airflow around the foot (Prydie and Hewitt, 2015). The author recommends to not leave the boot on for more than 3 hours at a time.
- **Cushy-Paw Slippers™** (www.therapaw.com/) are padded, grippy slippers for indoor comfort and protection.
- **Ruffwear boots—Grip trex™** (http://www.ruffwear.com/Barkn-Boots-Grip-Trex_3) are all-terrain boots and are a little thicker soled with canvas upper. They can be adapted for sore feet by lining with foam and can be used for multiple applications of indoor and outdoor foot protection.

- *Lewis Brand Vented Rubber Dog Boots™* (<http://www.gundogsupply.com/-950-.html>) protect the dog's feet from sand burrs, rocks, and brush, and can also protect an injured foot or sore pad. Because they are molded to foot shape they can help to keep the toes in proper extension for gait in animals that have mild proprioceptive issues. The boots are made of tire tread rubber with a tread on the sole to increase the life of the boot and for non-slip traction.
- *Foufoudog™* (www.foufoubrands.com) are rubber-dipped socks with an anti-slip silicone sole that claim to keep paws dry, clean, and warm! The stretchy sock design makes it easy to put on, and the hook and loop allows you to wrap the sock securely so they do not fall off. The strap can be detached for quick walks or use around the house. The innovative rubber dip protects paws from outdoor elements such as water, snow, dirt, salt, and hot pavement.
- *Dr. Buzby's ToeGrips™* (www.toegrips.com/) provide traction on slippery floors. These are rubber rings that fit around the dog's nail to provide grip through friction just below the nail tip. Care must be taken to pull the grips down from the cuticle as the nail grows. Dogs tolerate these well but note that some breeds, such as Dobermans, who do not have nail contact with the ground, and some individuals who do not engage the toenails when slipping, will not benefit from toe grips (Prydie and Hewitt, 2015).
- The purpose of the *Medipaw® X* boot (various suppliers) is to protect various appliances, such as leg bandages, splints, and casts. It can be used over incisions and is good for patients trying to bite or disturb a wound as in lick granulomas. The boots or suits are available for dogs and cats.

Slings and Harnesses

Slings may be strapped around the belly or fitted for the forelimbs, hindlimbs, or both. They should have long, hand-held straps attached to allow proper body mechanics to avoid personal injury to the handler when supporting the pet. Slings aid in transitioning a recumbent animal, especially larger dogs, to

a standing position (Adamson *et al.*, 2005). They can also assist with ambulation and prevent falls on slippery floors, especially after surgery, to avoid further injury to the animal. Support slings are available for forelimb assistance and patients with amputations. Patients who cross their pelvic limbs may benefit from the placement of a rolled cotton towel within a pelvic limb sling (Marcellin-Little and Levine, 2014). Slings used for the forelimbs should not obstruct respiration, and urine flow should not be compromised with hindlimb slings in male dogs. Slings should have a soft lining against the animal's skin to avoid irritation and sores, and they should be washable. Slings may be used for supported standing for therapeutic exercises such as repeated sit to stands. When documenting patient progress, the amount of assistance given through sling support can be rated as minimum, moderate, or maximal.

Harnesses are a more useful aid to mobility. They can be worn for several hours a day and provide easy access to a "handle" to lift the dog up (Prydie and Hewitt, 2015). This is particularly important if it is a larger dog to help the owners avoid injuring themselves. Animals with cervical injuries, especially following neurosurgery, will require chest harnesses to support their mobility without interfering with the injury site. There are also harnesses available with extra-long straps. These are useful for small dogs where the owners are likely to damage their back by reaching down to hold the harness.

Where do I get Slings and Harnesses?

- *GingerLead Sling™* (www.gingerlead.com/) is a sling with handles and an extension leash to the collar for control of patient speed and motion. It allows the patient to feel supported while providing controlled mobility.
- *FourFlags Quick Lift™* (<http://www.fourflags.com/s.nl/sc.11/category.6990/.f>) is a sling made of nylon pack cloth, available with or without fleece lining.
- *Komfy Fleece Dog Mobility Sling™* (<http://www.seniordoggy.com/komfy-fleece-mobility-sling/>) is a fleece-lined sling.

- *Hartmans Hip Helper™* (www.hartmanharness.com/) is a harness which is designed to be worn for hours at a time. It provides a handle over the rump for easy lifting assistance.
- *Help 'Em Up Dog Harness™* (<http://helpemup.com>) is a padded harness designed for daily wear. Handles sit over the withers and over the rump. Front harness and rear harness can work together or separately. This is M.E. Goldberg's favorite harness because it offers the most control.

Ramps and Stairs

Ramps for dogs are useful for getting in and out of a vehicle, getting into the owner's bed, or going up and down steps. They are especially useful in geriatric patients. Ramps offer a solution that helps to protect the dog's joints when jumping and the owner's back when lifting the dog in and out of a vehicle. It is also advisable to have a ramp for getting out of a vehicle if the dog has had front-leg surgery, is prone to carpal hyperextension injuries, or has a tendency to hyperextend its carpus, as jumping out of a car will aggravate this (Prydie and Hewitt, 2015). Telescoping ramps are available. Measure the height of the bed or vehicle to calculate how steep a ramp will be, based on its length.

Stairs are helpful for pets wanting to get to a higher elevation such as owner's beds, sofas, or window access. Cats particularly like using carpeted stairs. Be sure to use stairs that are strong enough for the weight of the pet and to look at pitch and overall width of the unit.

Where do I get Ramps and Stairs?

- *Discount Ramps™* (www.discountramps.com) have a wide variety of ramps.
- *Pet Classics Indoor Pet Ramps™* and *Pet Classics Outdoor Pet Ramps™* (www.petclassics.com/) are wooden ramps made for cats and dogs and covered with carpet.
- *Wayfair™* (<http://www.wayfair.com/Dog-Ramps-and-Stairs-C1804268.html>) offer a variety of pet ramps and stairs for both indoor and outdoor use.

- *Doctors Foster and Smith™* (<http://www.drsofostersmith.com/cat-supplies/furniture/ramps-and-stairs/ps/c/3261/10345/11278#>) have a range of cat ramps and stairs.
- *SolVit™* (<http://www.solvitproducts.com/ramps>) supply ramps and ramp accessories to help the pet get in a vehicle.

Carts and Wheelchairs

Carts can allow functional independence for impaired animals. They aid in ambulation and can be used to provide therapeutic exercise. They are normally suggested for paretic or paralyzed patients, patients with severe osteoarthritis, and obese patients, and may be used with other devices such as boots (Borghese *et al.*, 2013). Carts have also been used for patients with long-term balance (vestibular and cerebellar) issues (Pazzaglia and Molinari, 2016).

When determining the most appropriate cart, consider the size and weight of the dog or cat, the amount of support needed, the patient's residual strength and mobility, and the caretaker's physical abilities. Getting a patient into and out of a cart can be very difficult and may require a lot of lifting.

Fitting Cart fitting can be challenging, and rehabilitation veterinarians and their technicians or nurses should plan on spending at least 45 minutes making adjustments aimed at getting the best ground clearance, allowing motion of ambulatory limbs with minimal restrictions and balancing the weight of the cart (Pancotto, 2015). Clients should practice getting the pet into and out of the cart during the fitting. Most cart manufacturers will assist with adjustments if sent a picture of the patient in the cart. Important positional factors include a straight topline (not bowed or curved), the patient should have good ground contact in the normal ambulatory limbs and toe-touch in weak or paretic limbs, and the strap should clip just behind the scapula. Overweight dogs may need added belly bands.

Where Do I Get Carts?

- *Dewey's Wheelchairs for Dogs™* (<https://www.wheelchairsfordogs.com/home.html>) are custom-made to a pet's measurements.
- *Doggon' Wheels USA™* (www.doggon.com/) are custom-made to the pet's measurements, weight, and activity level. Dog and cat carts are available. The patient can be backed into the cart and the rear limb sling then snaps into straps on the cart.
- *Eddie's Wheels for Pets™* (<http://eddieswheels.com/>) are custom-built for dogs and cats. The welded padded saddle gives solid support but the pet does have to be lifted into the cart. For cat patients, they send out a cat harness for the cat to get used to wearing while the wheelchair is built. The cat harness should be put on the cat to wear at all times, so that he or she gets used to wearing something and once the cart arrives, a piece of double-sided Velcro is used to attach the cat harness to yoke of the cart.
- *K9 Carts™* (www.k9carts.com/) are adjustable and available for dogs and cats.
- *Walkin' Wheels* (www.handicappedpets.com) come in standard sizes that are adjustable. Different-sized wheels and bar lengths make a kit so that the cart can be modified for a patient. For larger dogs, steel reinforcement bars are added. These carts are better for rear limb issues only; a belly band can help patients with abdominal weakness.

Beds and Bedding

Orthopedic dog beds provide the necessary and supportive firmness a geriatric pet needs to sleep comfortably. These beds give a pet the support and comfort of orthopedic foam, making them the best pet beds for older, ailing, and postoperative dogs, or those with arthritis. A pet bed provides a sense of security. Advise the client to locate a comfortable place in their home for the pet bed, and keep the bed in the same location so the pet learns that he or she has a special resting spot. Pet beds are comfy and most pets would prefer to sleep somewhere soft that provides

cushioning and insulation instead of on the cold, hard floor.

Important Factors to Consider Before Choosing a Pet Bed

- *Sleeping style.* Some pets like to sleep curled up in a cozy ball, but others like to sprawl out. Have the owner observe the pet's sleeping style to help determine the right bed for him or her. Does the pet like the security of leaning against something? Or does the pet like to have enough room to stretch? These are important questions to consider before choosing.
- *Pet type, size, and activity level.* Is this for a dog or cat? Big dogs will need a larger, sturdier bed with a thick, more durable cover than a smaller dog would need. Cats prefer window perches to satisfy their curiosity. Older pets would benefit from heated pads or orthopedic beds for relief from arthritis.
- *Style and budget.* Nowadays, pet beds come in a wide range of styles, colors, materials, and prices, so there is something for every taste and budget.

Popular Types of Pet Beds

- *Orthopedic pet beds* provide extra cushioning for bones, helping to soothe painful pressure points. The large size of an orthopedic pet bed is great for pets that like to sprawl out while sleeping. The mattress type of pet bed is recommended for large dogs, while orthopedic window perches are recommended for cats. Some of the smaller orthopedic pet beds may be used for either dogs or cats.
- *Donut (bolster) beds.* If the pet likes to curl up or sleep with his or her head resting on a pillow, a donut or bolster bed is good choice. This style of pet bed usually has a cushioned bottom and a raised side. It is cozy for smaller pets since the round shape of the pet bed helps retain body heat.
- *Pillow or cushion beds.* This pet bed is basically a large pillow or cushion. It comes in many different sizes, and is a great choice for pets that like to stretch out while sleeping.

- *Heated pet beds* are recommended for older pets that may have joint pain or stiffness. The gentle heat soothes and reduces stiffness, keeping the pet warm and cozy while sleeping. Even smaller pets, such as cats, can benefit from a heated pet bed since they lose body heat easily. Consider a using a heated pet bed if the client lives in a colder climate.
- A *pet cot* should be considered if the dog or cat likes to rest outdoors. Pet cots protect the pet from the rough ground, especially asphalt that may get too hot in the summer or too cold in the fall or winter. Many pet cots are made of a waterproof fabric which helps make cleaning easier. The sturdy frame makes it ideal for larger dogs.

Where Do I Get Pet Beds?

- *Kuddle Kots™* (<http://kuddlekots.com/>) are made as ortho or chiropractic beds. They were originally designed for Dachshunds. It is best to speak to the manufacturer.
- *The PetCot™ Company* (www.petcot.com/) makes beds for home, boarding kennels, and veterinarians. They can be vinyl, making them easy to clean, or have a fleece cover. Cage inserts are available for the veterinarian's hospital and a raised pet bed for dogs with incontinence. Replacement parts, fleece covers, and orthopedic inserts are available as well.
- *The SleePee Time Bed®* (<http://www.handicappedpets.com/sleepee-time-bed-for-incontinent-pets>) is a new product specifically designed to help dog incontinence and older dogs get a good night's sleep. It can also be used simply as a cool, comfortable sleeping surface for any pet.
- *Care-A-Lot Pet Supply* (<http://www.carealotpets.com/departments/cat-supplies/cat-beds-and-cuddlers.aspx>) offer cat beds and cuddlers, including heated beds, plush beds, round beds, and one hanging cat condo.
- *Wayfair® Dog Beds* (<http://www.wayfair.com/Dog-Beds-C409475.html>) offer orthopedic beds, bolster beds, fleece beds, pet cots, and pet sofas.

Hydraulic Patient Lifts

Hydraulic lifts are either manual or electric. They can be found at medical supply stores and purchased new, used, or rented. A few are being made specifically for dogs. It is important to purchase a sling attachment used specifically for dogs even if you buy a human version lift. A lift can be used to manage a large tetraparetic or tetraplegic cervical patients. A patient lift (patient hoist, jack hoist, hydraulic lift) (Figure 10.4) is usually a sling lift, allowing the patient to assume a stand position. The patient may need to be supported by therapy balls.

- The *Equa-Lift Hoist™* (<http://eddieswheels.com/p/19/Hospital-Equipment>) is used in conjunction with a Hoyer™ lift or overhead track system and a Help 'em Up Harness™. The hoist provides a safe, balanced way to move disabled animals in and out of the therapy stand, clinic quad cart, or hydrotherapy unit.

Premade Orthoses

Splints and Braces

A splint is a device applied to any body part to help stabilize a structure, promote healing, protect against injury/re-injury, prevent/correct deformity, and/or assist with function. A splint is lightweight, easy to apply and remove, and adaptable. It can be static or dynamic, rigid or flexible and can be used as a coaptation device or solely for therapeutic purposes (Borghese *et al.*, 2013).

A brace is defined as a durable device used for immobilizing a body part. It renders the body part static, rigid, and fixed (Borghese *et al.*, 2013).

Splints The primary functions of a coaptation device include protection, absorption, compression, and stabilization.

- *Air splints* can be used to help a dog to stand while the veterinarian works on them, acting as an extra set of hands to stabilize a joint or limb. Air splints are waterproof and reusable, easy to apply, inflate,



Figure 10.4 Assisted standing with a Hoyer list and therapy ball.

deflate, remove, and clean, making them ideal for in-home use by savvy clients. When inflated, they conform to the animal's limb and are not associated with pressure sores. They are transparent and can be half filled with water, refrigerated, then used as cold packs over an entire limb. <http://www.therapaw.com/airpressure-splint.aspx>

- **Thermoplastic splints** are made of thermal plastic material heated in hot water until it becomes soft. Once soft, the material can be cut and molded into a custom shape for the patient's needs. After the splint is made, it needs to be cooled to become solid. This process can be speeded up by running cold tap water on it or by wrapping a cold wet crepe bandage over the splint. Hook and loop (Velcro) straps hold the splint in place, and padding/liner materials increase patient comfort. The problem with hook and loop is that it needs to be applied correctly to ensure fixation. If it is too loose, the splint will cause friction and abrasions may occur (Prydie and Hewitt, 2015). Thermoplastic sheets are widely used with human patients to make custom splints and braces that partially or

totally immobilize a body part. They have now been adapted for use with animal patients (Figure 10.5).

Padding materials are used with splints to protect bony prominences, areas of fragile skin, and less-protected areas of the body. Padding should only be used in the middle part of the splint and not end to end as this will significantly alter the fit of the splint particularly with a large amount of padding. The other consideration when using padding is that it can become wet and trap distasteful smells.

Braces Bracing is used to support and stabilize a body part. Some wraps and braces are not custom-made. These could be termed generic braces.

Cervical neurologic patients (Drum *et al.*, 2014) may need bracing made from thermoplastic or fiberglass cast material that extends from mid-sternum cranially to the chin. The brace prevents ventral flexion of the neck (Figure 10.6). It should be closely monitored for excessive tightness, areas of irritation, and wetness. Many patients will have difficulty eating, so syringe feedings with water and pureed food may be required. For more



Figure 10.5 Thermoplastic sheet for devices.

severely affected cases, surgery is often required to stabilize the joint between C1 and C2. After surgical stabilization, a brace is required for 4–6 weeks postoperatively.

- *L'il Back Bracer™* (www.lilbackbracer.com/) is an adjustable design for comfortable, stable support when dealing with intervertebral disc disease and other back pain in dogs. It provides comfortable but firm support from above and below to stabilize the dog's spinal column. It prevents further injury and relieves pain for dachshunds and other small-breed dog before or after surgery, sometimes entirely preventing surgery.
- *DogLeggs Therapeutic and Rehabilitative Products* (<http://www.dogleggs.com/index.cfm>) offer various braces. *DogLeggs' Carpal Support* are indicated for carpal hyperextension, carpal instability/sprain or strain injury, carpal osteoarthritis, immune-mediated joint disease, angular limb deformity, chondrodystrophic dogs in need of carpal support, post-op management of carpal arthrodesis or avulsion fractures. It is made from a closed-cell foam with synthetic rubber material and is laminated with a nylon jersey fabric. It is



Figure 10.6 Neck brace for an Irish Wolfhound.

lightweight yet durable and has incredible tensile strength. The product is machine washable. *DogLeggs' Tarsal Support* is indicated for mild to moderate Achilles tendon disruption, tarsal instability/sprain or strain injury, tarsal osteoarthritis, immune-mediated joint disease, angular limb deformity and chondrodystrophic dogs in need of tarsal support. It is constructed from a three-dimensional fleece-faced, highly breathable, four-way stretch textile with wicking properties. It allows air to circulate while keeping the joint warm. It contains padding to protect the bony prominences of the hock. Hook and loop fastening is used to secure the product. The product is machine washable. *DogLeggs' Shoulder Stabilization System™* is indicated for conservative management of medial shoulder instability, postoperative management of medial shoulder instability or shoulder reconstructions, conservative or postoperative management of certain scapular fractures, postoperative management of certain shoulder luxation repairs, and swimmer puppy syndrome. The system prevents abduction (moving away from the body) of the forelimbs and limits shoulder extension and flexion. The limited extension results in a mildly shortened gait when walking. The shoulder stabilization orthosis is made from breathable neoprene that can be worn continuously.

- *Thera-Paw Assistive and Rehabilitative Pet Products* (www.therapaw.com/) offer a variety of products. *Custom Carpal Wraps* are designed to provide customized support, stability, or immobilization. They come in three thicknesses. The same is true of the *Custom Tarsal Wrap*. The *Dorsi-Flex Assist™* (Prydie and Hewitt, 2015) can be used to good effect in dogs and cats that knuckle over on their back legs because of reduced neural input from degenerative myelopathy, peripheral nerve injury, or proprioceptive deficits. It can be used in dogs with weakness; however, care needs to be taken to ensure that the dog has enough strength to lift the foot with

the boot on and that the dog has enough stability around the hip to control the movement of the leg. The elastic straps are placed under tension and as the animal brings the leg forwards, the elastic aids the foot being moved up into dorsiflexion.

- *Biko™ Progressive Resistance Bands* (www.animotionproducts.com/) (Prydie and Hewitt, 2015) can aid gait where there is bilateral back leg weakness such as degenerative myelopathy, wobblers, or degenerative disc disease with mild to moderate ataxia. They consist of two leg cuffs that have elastic straps clipped onto them. These elastic straps attach onto a chest harness. There are different colors of elastic relating to progressive strengths of elastic and the device works by pulling the foot forwards as the elastic recoils after being stretched by extending the hips. Caution is needed when selecting clients to use this system on as it does not prevent knuckling and it is not appropriate for acute painful conditions. It is also not suitable for patients with unilateral problems as it can increase ataxia, or those with paralysis and no motor function in one or more legs.

Custom Orthotics

Custom orthoses are made from casts of the patient's limb or are fabricated directly on the patient using a moldable thermoplastic polymer (Marcellin-Little *et al.*, 2015). These can be designed for aquatic therapy or to be worn for playing in water. The most common orthoses are used to restrict the excessive joint motion present in patients with ligamentous or tendinous problems, including patients with ruptures of their palmar fibrocartilage or common calcaneal tendon (Marcellin-Little and Levine, 2014). Custom fabrication becomes essential when a standard or semi-custom product (Borghese *et al.*, 2013):

- is not available in the correct size,
- does not meet the specific needs of the patient,

- cannot be altered to meet the patient's needs,
- incorporates a joint or joints not affected by the condition, or
- simply does not exist.

Orthoses may be hinged to enhance their functionality (Levine and Fitch, 2003). Orthoses with passive hinges may enhance limb function because part or all of the motion of the tarsus (or carpus) may be maintained. Orthoses may have dynamic hinges that may be used to enhance the support of weak joints or to stretch contracted joints. Hinges are made dynamic by using springs embedded in the hinges or elastic bands (Marcellin-Little and Levine, 2014) (Figure 10.7).

A custom carpal orthotic would basically eliminate the need for cage rest while tissues are healing but, in contrast to complete immobilization of the carpus, allows some amount of tissue loading and can be removed by an owner to enable provision of passive therapeutic joint motion and daily inspection of the skin for abrasions (Tomlinson and Manfredi, 2014).

Custom orthotics have been used in conjunction with autologous mesenchymal stem

cell treatment, as in the case of a 4-year-old Border Collie where the owner elected conservative treatment over surgery. The owner was thoroughly informed about the experimental nature of this method of treatment (Case *et al.*, 2013).

Where Do I Go?

A professional company such as OrthoPets™ (www.orthopets.com/) needs to be employed to make custom orthotics. Examination of the patient, measuring, and continued communication between the rehabilitation veterinarian and the orthotics company helps to ensure custom fit of the device. Courses are available to enable the veterinarian and their rehabilitation veterinary technician to learn how to create a fiberglass mold and send the required materials to the company (http://www.caninerehabinstitute.com/Orthotics_Prosthetics.lasso).

Other companies are:

- *Animal Ortho Care, LLC* (www.animalorthocare.com/)
- *K-9 Orthotics and Prosthetics Inc.* (www.k9orthotics.com/)
- *Handicapped Pets.com* (<http://www.handicappedpets.com/k9-dog-orthotics-prosthetics>)
- *My Pet's Brace* (www.mypetsbrace.com/)

Types of Custom Orthotics

- Shoulder orthoses
- Elbow orthoses
- Antebrachial and carpal orthoses
- Hip orthoses
- Stifle orthoses
- Hock orthoses
- Distal limb orthoses (distal to carpus or tarsus)
- Spinal orthoses.

Custom Prosthetics

Not long ago, amputation of a limb for a small animal patient was quite normal when a salvage procedure was needed from trauma



Figure 10.7 Rear, right leg orthosis.



Figure 10.8 Front, left leg prosthetic, lateral view. *Source:* Courtesy of Dr. Douglas Stramel.

or disease. It is now being understood that the consequences of that missing limb or limb segment are not good. Organizations advocating pain management such as the American College of Veterinary Anesthesia and Analgesia, the International Veterinary Academy of Pain Management, the American College of Veterinary Sports Medicine and Rehabilitation and the American Association of Rehabilitation Veterinarians are understanding the importance of biomechanics in normal quadruped locomotion. If a total or partial limb is missing then there can arise limited mobility and endurance, increased metabolic demand, weight gain, support limb breakdown, chronic neck and back pain, and premature euthanasia (Mich, 2014). The re-establishment of quadruped function and structure has been made possible now in veterinary medicine. The end goal is to provide a limb that allows as close to normal ambulation as possible.

Types of Prosthesis

There are two types of prosthetic limbs available: the socket design and the ITAP (intraosseous transcutaneous amputation prosthesis). Socket-based prostheses provide a socket for the residual limb to reside in. Usually an extension (in the form of a



Figure 10.9 Front left, leg prosthetic anterior view. *Source:* Courtesy of Dr. Douglas Stramel.

foot or paw) is provided for the limb to make contact with the ground (Figures 10.8–10.10). The socket prosthesis requires owner care and must be put on or removed from the patient.



Figure 10.10 Walking with a front left leg prosthetic, anterior view. *Source:* Courtesy of Dr. Douglas Stramel.

Contraindications for socket prostheses may be linked to the owner, the patient, or the medical condition (Marcellin-Little *et al.*, 2015). Owner-related contraindications include a potential lack of interest, motivation, supervision, or financial ability to get involved. Patient-related contraindi-

cations include being difficult to handle because of an aggressive personality (Box 10.1).

ITAP provides an implanted endoprosthesis (into the bone) to which an exoprosthesis (external prosthesis) is attached (Mich, 2014). The clear advantage of an ITAP is direct skeletal integration of the exoprosthesis (Fitzpatrick *et al.*, 2011), and it has been shown that biological integration of osseous and dermal tissues with ITAP is reliable and robust. The tremendous variability in veterinary patients requires adaptability in socket design, components, and prosthetic limb mechanics to accommodate differences in the degree of injury, body type and condition, species, breed, size, lifestyle, sport or activity, and terrain (Kaufmann and Mich, 2014) (Box 10.2).

Examples of Successful ITAP Procedures

Some procedures in which ITAP has been used successfully include:

- total hip replacement (Liska *et al.*, 2009; Vezzoni *et al.*, 2015),
- total knee replacement (Mann *et al.*, 2012),
- canine elbow arthroplasty (Burton *et al.*, 2013), and
- titanium partial limb prosthesis in a white crane (Rush *et al.*, 2012).

Box 10.1 Contraindications for prosthetic devices in cats and dogs	
Owner • Lack of interest • Motivation • Supervision • Financial ability to get involved	Medical condition • Highly mobile skin in relation to underlying tissues • Neurologic deficits • Compromised joint mobility • Local neoplasia • Infection • Stump pain
Patient • Difficult to handle because of an aggressive personality	<i>Source:</i> Marcellin-Little <i>et al.</i> (2015). Reproduced with permission of Elsevier.

Box 10.2 Orthopedic conditions that will adapt to veterinary prosthetic devices**Thoracic limb prosthetics**

- Subtotal mid-shaft radius or ulna amputation (40% antebrachium retention required)
- Subtotal radiocarpal disarticulation
- Subtotal intercarpal disarticulation
- Subtotal carpometacarpal disarticulation
- Amelia—a birth defect of lacking one or more limbs
- Congenital limb derangements
- Traumatic limb amputation

Pelvic limb prosthetics

- Subtotal mid-shaft tibia or fibula amputation (40% crus retention required)
- Subtotal tarsocrural disarticulation
- Subtotal level intertarsal disarticulation
- Subtotal level tarsometatarsal disarticulation
- Amelia—a birth defect of lacking one or more limbs
- Congenital limb derangements
- Traumatic limb amputation

Source: Adapted from Kaufmann and Mich (2014).

Follow-Up Care for the Prosthetic Patient

It is critical for the veterinary prosthesis patient to undergo physical rehabilitation. The best way to ensure the highest level of success with a prosthetic device is to follow a rehabilitation plan established by a certified canine rehabilitation professional (CCRT or CCRP). Sensory feedback extends from the top down rather than from the ground up. Through rehabilitation, the prosthesis patient relearns proprioception, balance, gaiting at different speeds, and ambulation over varied terrain (Mich, 2014). Training companion animals to use socket prostheses is similar to managing limb disuse. It relies on habituating patients to tolerate the exoprosthesis at rest and loading it (using it) when standing, when walking slowly (indoors), when walking more rapidly (outdoors), when trotting, when galloping, and then during other activities of daily living (e.g., climbing and walking down steps, climbing and walking down stairs, jumping up, jumping down, playing) (Marcellin-Little *et al.*, 2015). Land-based therapeutic exercise is essential. Balance, proprioception, muscle timing (neuromuscular retraining), and coordination lay the foundation for proper device use (Mich, 2014). These must be mastered on land so that the patient can

learn response to normal ground reaction forces and shifts in their total body force vectors (Mich, 2014).

Veterinary orthoses and prostheses are considered durable medical devices. As such, they should never be prescribed or dispensed without client training and a comprehensive follow-up plan. Veterinary orthotic/prosthetic patients should be assessed at least annually (Figures 10.11–10.13). Device adjustment and refurbishment are expected in order to continue meeting therapeutic goals (Borghese *et al.*, 2013).

It is beyond the scope of this chapter to discuss the complete process of the veterinary evaluation, defining therapeutic goals, measuring for the mold cast to be sent to the prosthetic fabricator, communication with the fabricator, delivery of the device, and ensuring proper fit. Manufacturing requires skilled modification of the model by hand or using computer-assisted design to build reliefs, which accommodate limb topography and create appropriate corrective forces when the completed device is applied to the limb. The modified model is the structure on which a thermoplastic shell is vacuum formed. The shell is then hand cut, trimmed, and ground to the final shape. Materials used to pad and line the shell vary. Hinges, straps, pads, and motion-limiting components complete



Figure 10.11 Double limb prosthetics for hindlimbs, lateral view. *Source:* Courtesy of Dr. Douglas Stramel.



Figure 10.12 Double limb prosthetics for hindlimbs, rear view. *Source:* Courtesy of Dr. Douglas Stramel.



Figure 10.13 Double rear limb prosthetics during exercise. *Source:* Courtesy of Dr. Douglas Stramel.

fabrication. The typical custom veterinary orthotic/prosthetic device cost varies with components and materials, but averages US\$600–\$US1000. This does not include the necessary appointments to ensure proper fit and function along with client education (Mich, 2014).

Where Do I Go?

Professional companies that fabricate these devices are listed at www.animaloandp.com/facilities.asp. Companies that are highly recommended include:

- *OrthoPets™* (www.orthopets.com/)
- *Animal Ortho Care, LLC* (www.animalorthocare.com)
- *Advanced Care Veterinary Services* (www.stoppetpain.com)
- *Dogs in Motion Canine Rehabilitation; OrthoPets Australia* (www.dogsinmotion.com.au)
- *Equisport Medicine; OrthoPets Washington* (www.equisportmedicine.com)
- *K-9 Orthotics & Prosthetics* (www.k-9orthotics.com)
- *A Loyal Companion; OrthoPets Arizona* (www.aloyalcompanion.com)
- *OrthoPets Europe; Elmtree House* (www.orthopets.co.uk)

- *Pawsitive Steps Rehabilitation Clinic; OrthoPets Michigan* (www.pawsitivestepsrehab.com)
- *Paws on the Go; OrthoPets Southern California* (www.pawsonthego.net)
- *Twin Cities Animal Rehabilitation Clinic; OrthoPets Twin Cities Minnesota* (www.tcrehab.com)
- *Veterinary Orthopedic and Sports Medicine Group (VOSM)* (www.vetsportsmedicine.com)
- *A Well Adjusted Pet; OrthoPets San Francisco* (www.awelladjustedpet.com)
- *Wheat Ridge Veterinary Specialists* (www.wheatridgeanimal.com)

Summary

The use of assistive devices, orthotics, and prosthetics has enabled many animals to continue to enjoy a high quality of life. Patients that might otherwise be at risk for euthanasia can return to an active lifestyle, reducing the risk of obesity and its associated comorbidities. Secondary or compensatory pain can be minimized by correcting or improving gait mechanics and re-establishing quadruped locomotion.

References

- Adamson C, Kaufmann M, Levine D, *et al.* (2005) Assistive devices, orthotics, and prosthetics, *Vet Clin North Am Small Anim Pract* 35:1441–1451.
- Bluebird Care (2015) *Assistive Technology from Ancient to Modern Times*. <http://bluebirdcare.ie/assistive-technology-ancient-modern-times/> (accessed January 3, 2015).
- Borghese I, Fair L, Kaufmann M, and Mich PM (2013) Assistive devices, orthotics, prosthetics, and bandaging. In *Canine Sports Medicine and Rehabilitation* (eds. MC Zink and JB Van Dyke). John Wiley & Sons, Inc., Ames, IA, pp. 201–222.
- Burton NJ, Ellis RJ, Burton KJ, *et al.* (2013) An ex vivo investigation of the effect of the TATE canine elbow arthroplasty system on kinematics of the elbow. *J Small Anim Pract* 54:240–247.
- Case JB, Palmer R, Valdes-Martinez A, *et al.* (2013) Gastrocnemius tendon strain in a dog treated with autologous mesenchymal stem cells and a custom orthosis. *Vet Surg* 42:355–360.
- Drum M (2010) Physical rehabilitation of the canine neurologic patient. *Vet Clin Small Anim* 40:181–193.
- Drum M, Werbe B, McLucas K, and Millis D (2014) Nursing care of the rehabilitation

- patient. In *Canine Rehabilitation and Physical Therapy*, 2nd edn. (eds. D Millis and D Levine). Elsevier/Saunders, Philadelphia, PA, pp. 277–304.
- Fitzpatrick N, Smith TJ, Pendegrass CJ, *et al.* (2011) Intraosseous transcutaneous amputation prosthesis (ITAP) for limb salvage in 4 dogs. *Vet Surg* 40:909–925.
- Hardie RJ and Lewallen JT (2013) Use of a custom orthotic boot for management of distal extremity and pad wounds in three dogs. *Vet Surg* 42:678–682.
- Kaufmann MW and Mich PM (2014) Custom external coaptation as a pain management tool: Veterinary orthotics and prosthetics. In *Pain Management in Veterinary Practice* (eds. CM Egger, L Love, and T Doherty). John Wiley & Sons, Inc., Ames, IA, pp. 155–160.
- Levine JM and Fitch RB (2003) Use of an ankle-foot orthosis in a dog with traumatic sciatic neuropathy, *J Small Anim Pract* 44:236–238.
- Liska WD, Doyle N, Marcellin-Little DJ, and Osborne JA (2009) Total hip replacement in three cats: surgical technique, short-term outcome and comparison to femoral head ostectomy. *Vet Comp Orthop Traumatol* 22:505–510.
- Mann KA, Miller MA, Khorasani M, *et al.* (2012) The dog as a preclinical model to evaluate interface morphology and micro-motion in cemented total knee replacement. *Vet Comp Orthop Traumatol* 25:1–10.
- Marcellin-Little DJ and Levine D (2014) Devices for ambulation assistance in companion animals. In *Canine Rehabilitation and Physical Therapy*, 2nd edn. (eds. D Millis and D Levine). Elsevier/Saunders, Philadelphia, PA, pp. 305–311.
- Marcellin-Little D J, Drum M, Levine D, and McDonald SS (2015) Orthoses and exoprostheses for companion animals. *Vet Clin Small Anim* 45:167–183.
- McDonald JW and Sadowsky C (2002) Spinal-cord injury. *Lancet* 359:417–425.
- Mich PM (2014) Topical review: The emerging role of veterinary orthotics and prosthetics (V-OP) in small animal rehabilitation and pain management. *Top Companion Anim Med* 29:10–19.
- Nicolson A, Moir L, and Millsteed J (2012) Impact of assistive technology on family caregivers of children with physical disabilities: a systematic review. *Disabil Rehabil Assist Technol* 7(5):345–349.
- Norton K (2007) A brief history of prosthetics. *In Motion* 17(7):11–13.
- Olby N (2010) Patients with neurologic disorders. In *BSAVA Manual of Canine and Feline Rehabilitation, Supportive and Palliative Care*. British Small Animal Veterinary Association. Gloucester, UK, pp. 168–170.
- Pancotto TE (2015) Canine wheelchair equivalents. *Clinician's Brief*, January 2015. <http://www.cliniciansbrief.com/article/canine-wheelchair-equivalents> (accessed January 7, 2016).
- Pazzaglia M and Molinari M (2016) Review: The embodiment of assistive devices – from wheelchair to exoskeleton. *Phys Life Rev* 16:163–175.
- Prydie D and Hewitt I (2015) Splints supports and aids. In *Practical Physiotherapy for Small Animal Practice*. Wiley/Blackwell, Oxford, pp. 195–209.
- Rush EM, Turner TM, Montgomery R, *et al.* (2012) Implantation of a titanium partial limb prosthesis in a white-naped crane (*Grus vipio*). *J Avian Med Surg* 26(3):167–175.
- Tomlinson JE and Manfredi JM (2014) Evaluation of application of a carpal brace as a treatment for carpal ligament instability in dogs: 14 cases (2008–2011). *J Am Vet Med Assoc* 244(4):438–443.
- Vezzoni L, Vezzoni A, and Boudrieau RJ (2015) Long-term outcome of Zürich cementless total hip arthroplasty in 439 cases. *Vet Surg* 44:921–929.

11

The Disabled Patient Part 2: The Neurological Patient

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Introduction

A brief review of neurological anatomy is in order. There are three main components to the nervous system: the central nervous system (CNS), the peripheral nervous system (PNS), and the autonomic nervous system (ANS). The PNS consists of the nerves and ganglia located outside the brain and spinal cord. The myelin sheath surrounding the peripheral axons is formed by Schwann cells, whereas CNS axons are myelinated by

oligodendrocytes (Thomson and Hahn, 2012). Dorsal and ventral nerve roots attach on each side of the spinal cord and carry sensory and motor axons. Lateral to the spinal cord, the dorsal and ventral roots fuse to form mixed, spinal nerves. Adjacent spinal nerves may fuse, in a plexus, to form named nerves in the periphery. There are 12 pairs of cranial nerves that innervate the head and extend into the body. Areas of sensory innervation of the skin are categorized as dermatomes, cutaneous zones, and autonomous zones.

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Somatic lower motor neurons innervating striated muscle of the body have their cell bodies sited in the CNS. Their axons travel in the PNS to connect to the muscle at the neuromuscular junction. A motor unit comprises a single lower motor neuron (LMN) and the group of muscle fibers it innervates.

Healing of Nerves

- Nerves are sensitive to oxygen and the glucose level of tissues.
- If conditions are optimal then nerves will regenerate axons. We aim to make conditions optimal with rehabilitation therapy.
- More severe injuries first undergo Wallerian degeneration. This is a degenerative process that occurs when a nerve fiber is cut or crushed. Part of the axon is separated from the nerve cell body and degenerates distal to the injury (Coleman and Freeman, 2010).
- Wallerian degeneration starts 24–36 hours after injury (a patient can lose more nerve function after the initial injury as the axons degenerate).
- Regeneration is slower in the CNS than the PNS. The myelin sheath in the PNS guides axon regeneration.
- In more severe nerve injuries where endoneurial tubes are disrupted, regenerating axons are no longer confined to their original myelin sheaths. They can meander into surrounding tissue, thus failing to reinnervate their proper end organs.
- Aging has been shown to retard the rate of axonal regrowth.

The Neurological Examination

The objectives in the management of an animal with a problem that may be related to the nervous system are to (Lorenz *et al.*, 2011a):

- 1) confirm that the problem is caused by a lesion in the nervous system;
- 2) localize the lesion in the nervous system;
- 3) estimate the severity and extent of the lesion in the nervous system;
- 4) determine the cause or the pathologic process or both;
- 5) estimate the prognosis with no treatment or with various alternative methods of treatment.

Taking a History

First steps in diagnosis of the neurological problem is taking a history. History can be taken by questionnaire, but a verbal history should be added. History should include length of the disease process/problem, current medications, prior and current treatments/therapies, and goals. All questions should be framed so that the answer “I don’t know” is an acceptable alternative; otherwise the client may hypothesize rather than relate facts (Lorenz *et al.*, 2011a). The technician/nurse will assist the veterinarian in history taking. Most likely, the neurological patient that presents for physical rehabilitation will have been previously diagnosed at their primary care veterinary hospital or specialist.

The client should be asked for an accurate list of current medications or supplements, chronic or recurrent illnesses, and, most important, the level of performance to which the animal is expected or needed to return (Millis and Mankin, 2014). A detailed description of the housing arrangement, including any unique features of the environment and information about the pet’s daily routine, is helpful in determining the necessary skill set for a patient. This description should include types of flooring, presence of obstacles such as stairs or pet doors, feeding and exercise schedules, and any other relevant details (Sims *et al.*, 2015). The patient’s current ability to function in the home should be determined with thorough questioning (e.g., Can they stand to eat? Is the food bowl raised? Does the surface they stand on to eat have adequate traction?).

Basic Neurological Exam

The physical rehabilitation veterinarian along with their rehabilitation technician/nurse will conduct a thorough neurological examination. The neurologic examination should always be performed in a calm, quiet place with minimal distractions. The aim of the neurologic rehabilitation examination is to assess function (and how that will affect home activities), define current limitations, assess pain, and define goals for treatment. Surfaces of different traction and small obstacles (cavalettis, low step, etc.) should be available to test stability and function. The rehabilitation veterinarian performs a neurological examination to document the current neurological status and become familiar with the patient's responses to measure progress throughout therapy. Sensation, motor function (including ability to stand and support weight and transition between postures), and bowel/bladder function are noted. The technician/nurse will document all findings in the medical record.

The general appearance, behavior, posture, and voluntary movements should be documented while the animal can freely move around the examination room. Interaction with the client, behavior that is normal, aggressive, excited, or apathetic, and a change in mentation should be documented because this may indicate cerebral/thalamic localization. Abnormal postures such as head tilts, head turns, kyphosis, or scoliosis are noted. Assessing posture includes transitions.

When evaluating the gait, we first determine if it is normal or abnormal. If it is abnormal, then we describe the patient as ambulatory versus non-ambulatory. From this point, further descriptive terminology will let us know the degree of abnormality. Paretic animals have decreased voluntary movement. The animals can be described as ambulatory, para/tetraparetic or non-ambulatory, para/tetraparetic. Tetraparesis is paresis in all four limbs. Hemiparesis is on one side or the body and paraparesis is of the pelvic

limbs. If there is no voluntary movement, the patient is described as plegic.

Baseline objective data such as thigh girth, goniometry, body weight, body condition scoring, stance analysis, force plate, and kinematic analysis (when available) may also be collected and recorded in the patient history during this initial assessment (Sims *et al.*, 2015).

Neurological Examination Overview

The factors that should be assessed in a neurological examination are as follows (Rylander, 2013):

- 1) Mentation
- 2) Posture and gait
- 3) Cranial nerves
- 4) Postural reactions
- 5) Spinal reflexes
- 6) Pain on spinal palpation
- 7) Pain perception.

An assessment of the patient's *mentation* should be made and recorded. This includes their mental attitude and response to the immediate environment. The client is the best judge of subtle changes in the patient's behavior in the home normal environment. Be sure to explore this issue when you obtain the history (de Lahunta and Glass, 2009) (Box 11.1).

Gait is evaluated if the animal can walk. A video of the patient performing a standard set of tasks (walking over ground, on a treadmill or underwater treadmill, trotting, circling, weaving, and stepping over obstacles) is often the most useful and accurate method for recording gait abnormalities and comparing over time (Olby *et al.*, 2014; Sims *et al.*, 2015). The grade of ataxia (1–4), limb ground clearance, circumduction, dorsigrade placement of the foot (knuckling), ability of an individual limb to support weight in stance phase, or any other abnormal limb motion is noted. Remember to assess the gait using support as necessary to prevent falls. These examinations are performed by the rehabilitation

Box 11.1 Neurological Examination Form

Created by Julia E. Tomlinson BVSc MS PhD DACVS CCRP CVSMT Diplomate, American College of Veterinary Sports Medicine and Rehabilitation and Mary Ellen Goldberg BS, LVT, CVT, SRA CCRA, CVPP, VTA-lab animal and physical rehabilitation

Posture, transitions, strength tests

Basic posture, ability to hold posture, control of postural transitions. Include degree of assistance needed to complete any transition

- a) Standing (e.g., kyphosis, lordosis, scoliosis, abduction of rear limbs). Rear sink into flexion in 10 seconds. Head tilt or turn.
- b) Sitting posture: ability to hold sit, rear limb position (e.g., extended or abducted), position of spine (e.g., back arched).
- c) Stand–sit–stand (e.g., descends controlled for 50% then falls into sit. When standing, pulls with front limbs, head down, needs 3–4 steps forward to gain full rear limb extension).
- d) Side laying–sternal: Left and right. Include any limitations in flexibility.
- e) Response to perturbations. For example:
 - i) Head motion while standing
 - ii) Leg lifts
 - iii) Side challenge/sway test
 - iv) Hop test.
- f) Range of motion of spine and limbs, goniometry of limb joints.
- g) Voluntary motions.
 - i) Cervical – extension, lateral flexion, ventroflexion, rotation
 - ii) Thoracolumbar – lateral flexion, extension, ventroflexion
 - iii) Squat/sit with full flexion of lower lumbar spine/sacrum.
- h) Gait
 - i) Grade of ataxia
 - ii) Ground clearance of limbs – flat surface, small step or obstacle
 - iii) Number of steps before falling (if applicable)
 - iv) Knuckling (e.g., intermittent or constant).
- i) Cranial nerve assessment – I–XII.
- j) Eye position, motion, tracking, menace, palpebral, pupillary light reflex direct and consensual, swallowing, cough reflex to tracheal pinch, cervical muscle tone.
- k) Myotactic reflexes – patella, sciatic, tibial, fibular (peroneal), biceps, triceps, extensor carpi radialis, etc. reduced, normal, increased, clonus.
- l) Note tremors, resting or when increase weight bearing load (e.g., front leg lifted, rear legs tremor)

Neurological examination form from the American Veterinary Medical Association website**Observation**

Mental	Alert	Depressed	Disoriented	Stupor	Coma
Posture	Normal	Head tilt	Tremor	Falling	Circling
Gait	Normal	Ataxia	Pelvic Limbs	All 4	
Paresis	Pelvic Limbs	Tetra	Hemi	Mono	
Other					

Postural reactions

Key: 4=exaggerated, clonus; 3=exaggerated; 2=normal; 1=diminished; 0=none; NE=not evaluated.

	LF	RF	LR	RR
Wheelbarrow				
Hopping				
EXT postural thrust				
Proprioceptive pos				
Hemistand/walk				
Placing – tactile				
Placing – visual				

Spinal reflexes

	LF	RF	LR	RR
Quadriceps				
Extensor carpi				
Flexion				
Crossed extensor				
Perineal				

Cranial Nerves

	L	R		L	R	Comments
II, VII – Vision menace			VIII – Nystagmus, resting			
II, III – Pupils resting			VIII – Nystagmus, change			
Stim L			V – Sensation			
Stim R			VII – Facial mm			
II – Fundus			V, VII – Palpebral flex			
III, IV, VI – Strabismus, resting			IX, X – Gag			
III, IV, VI, VIII – Strabismus, position			XII – Tongue			

Sensation (Locate and describe abnormal)

Hyperesthesia		
Superficial pain		
Cutaneous reflex		
Deep pain		

What is the problem? Where is the lesion? What are the most probable causes of this problem?
What is your plan to establish a diagnosis?

Assessment

Anatomic diagnosis

Problem	Rule out location

Likely location of one lesion

Source: Adapted from the standard neurological evaluation format used by the *Journal of the American Veterinary Medical Association* in the featured title: "What is your Neurological Diagnosis?"

veterinarian while the veterinary technician/nurse is recording or documenting the exam.

A *cranial nerve* examination should be done, evaluating cranial nerves 1–12 systematically.

Postural reactions are tested. These include (Lorenz *et al.*, 2011a):

- proprioceptive positioning,
- wheelbarrowing,
- hopping,
- extensor postural thrust,
- hemistanding and hemiwalking (Figure 11.1),
- placing (tactile),
- placing (visual),
- sway test, and
- tonic neck (the tonic neck reaction involves extension of the head and neck so that the nose is directed dorsally).

Spinal reflexes, including myotatic reflexes and flexor reflexes, should be evaluated.

The neurologic examination also includes *palpation for sensation* and particularly for areas that are painful. This is not limited to the muscles and joints of the spine but includes the whole body. The skin dermatomes are tested where applicable using a fabric pinwheel or the end of a hemostat (charts can be found in veterinary neurology texts).

In general, the ability to perceive *pain* is only assessed in patients with loss of motor function; however, young patients presenting with signs of a sensory neuropathy are an exception (Rylander, 2013). Muscles are palpated for tone both resting and in supporting weight where applicable. Muscle atrophy is noted, along with the innervation of that muscle (for example atrophy of sciatic innervated muscles of the left rear is noted).

The temperament of the patient often dictates which treatments and exercises are realistic. If a patient is aggressive or



Figure 11.1 Hemistanding. The therapist uses their legs to support the trunk if needed. Hemiwalking is toward the therapist, support as needed. Level of support is noted.

excessively fearful, it may be difficult for the client or veterinary professional to perform complex exercises (Drum, 2010). Low-stress handling should be used where possible.

Assessment of Transitions

As the patient progresses, the ability to transition, evaluating patient strength and

ability to rise independently, should be documented on the form (Figure 11.2). A patient may appear to be incapable of getting up from a down position but with only light support may be able to make the transition, or a patient may need almost full body support to complete the task (Panko, 2014). These details should be noted on the examination form (Figures 11.3, 11.4, and 11.5).

Figure 11.2 Cookie stretch to aid transition. Voluntary movement for a treat is encouraged. Assistance is given as needed and level of assistance noted.



Figure 11.3 Lateral to sternal transition as demonstrated by Jenn Panko. *Source:* Courtesy of NAVTA Journal.

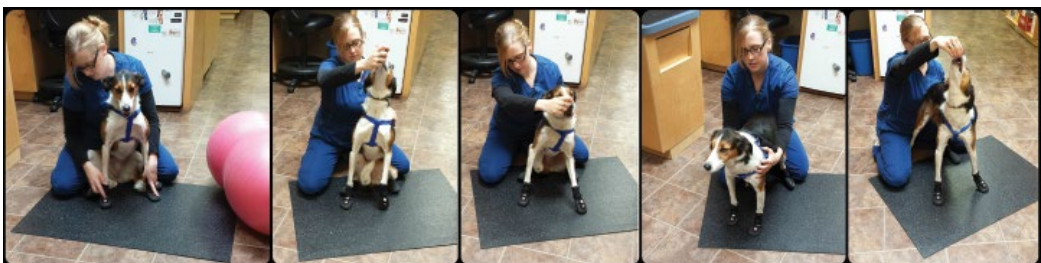


Figure 11.4 Sternal to sit transition as demonstrated by Jenn Panko. *Source:* Courtesy of NAVTA Journal.



Figure 11.5 Sit-to-stand transition as demonstrated by Jenn Panko. Source: Courtesy of NAVTA Journal.

Examples of transitions are:

- *Lateral to sternal recumbency*—Guiding the patient into a sternal position will enable it to strengthen muscles used for this task.
- *Sternal recumbency to sit*—Assist them with stifle flexion and with limb positioning and correct foot placement as required.
- *Sit to stand*—From a sit with stifles in flexion and appropriate support, assist patient to stand and back down into a sit; guide only as needed.
- *Stand to assisted walking*—Assisted sling walking is a great way to provide patients with safe ambulation and weight bearing.

Evidence-Based Information About Neurologic Rehabilitation

Looking at the human literature, conditions that require neurological physical rehabilitation include stroke, traumatic brain injury (TBI), spinal cord injury, and Guillain-Barré syndrome (Jorge *et al.*, 2014). Therapy has been proven beneficial and effective for some of these patients (Brody, 1999).

Neurological diseases present unique circumstances in which veterinary rehabilitation therapy has a critical role in maintenance and recovery of function. Dysfunction of the nervous system can cause loss of motor and autonomic function and a range of sensory abnormalities, including loss of sensation (analgesia), abnormal sensations (paresthesia), and heightened sensitivity to stimuli (hyperes-

thesia) (Olby *et al.*, 2005). Papers published in the veterinary literature support the usefulness of rehabilitation in recovery from neurological injury and non-surgical management of neurological conditions (Drum, 2010). Reports of case management with rehabilitation in the veterinary literature include prolonging survival time for dogs with degenerative myelopathy (Kathmann *et al.*, 2006), having a positive influence on outcomes of dogs with fibrocartilaginous embolism (Gandini *et al.*, 2003), and improving time to return to function in patients with intervertebral disc disease using laser therapy after decompressive surgery (Draper *et al.*, 2012). In addition, electroacupuncture was shown to be more successful than decompressive surgery in a small study of dogs with longstanding severe neurologic deficits from intervertebral disc disease (Joaquim *et al.*, 2010).

As part of the veterinary rehabilitation team, the credentialed veterinary technician under the supervision and direction of a licensed credentialed rehabilitation veterinarian is an integral part of caring for hospitalized recumbent and/or neurological patients. Physical rehabilitation during recovery from neurological disorders is important not only for strengthening, gait retraining, and increasing flexibility, but also for pain reduction and improvement in quality of life (Lorenz *et al.*, 2011b). Understanding the potential complications and risks, and implementing strategies to minimize these, can reduce the duration of hospitalization, improve patient comfort, and promote faster return to function.

Nursing Care of Neurological Patients in Clinic

Neurological patients often present as recumbent. This is usually patients with severe spinal or generalized LMN disorders (Olby, 2010). Recumbency can cause secondary problems associated with the respiratory system, bladder and bowel function, skin and musculoskeletal system, and can affect the patient's ability to eat and drink and its attitude. The principles of managing such patients include (Olby, 2010):

- awareness of potential complications associated with recumbency;
- regular careful assessment of the patient;
- taking appropriate preventative steps; and
- early and aggressive treatment of developing complications.

Bedding for the recumbent patient should be soft to avoid decubital ulcers. Turning frequently lowers the risk of ulcers and decreases the likelihood of complications such as hypostatic pneumonia or atelectasis (Calvo, 2012).

Ambulatory patients presenting with neurological signs characterized by incoordination (ataxia) or weakness (paresis) have the potential to progress to recumbency. As these patients are not steady on their feet, providing high traction surfaces for them to move around on is essential. The top five care tips for recumbent dogs are listed in Box 11.2.

Box 11.2 Top five care tips for down dogs

- 1) Provide proper bedding
- 2) Teach bladder management to the client
- 3) Rotate, massage and provide basic therapies
- 4) Know nutritional needs
- 5) Monitor for pain and neurologic changes

Source: Adapted from Weber (2016).

Respiratory System

Recumbency on its own can lead to secondary complications, including atelectasis and aspiration pneumonia, which are independent from the disease process causing recumbency. Hypoventilation can occur when a neurological disease process is severe enough to cause recumbency in all four limbs (e.g., cervical myelopathy or a severe encephalopathy). The respiration centers of the brainstem can be affected. Patients with generalized LMN diseases affecting the laryngeal and pharyngeal muscles and the esophagus (e.g., myasthenia gravis) are particularly predisposed to aspiration pneumonia. Preventative nursing care is crucial to the outcome for these patients (Calvo, 2012).

Assessment of respiration includes the following:

- Regular assessment and recording of the respiratory pattern and rate up to every 4–6 hours in severely affected patients; less often in more stable patients.
- If there is a suspicion of aspiration pneumonia, rectal temperature should be taken at least twice daily to monitor for pyrexia.

Measures to prevent respiratory complications include the following:

- Regular turning of the patient (every 4–6 hours), adopting a sternal position as often as possible using appropriate padding (record each time the position was changed: e.g., from sternal to left lateral to sternal to right lateral to sternal).
- Offer water and food only when the patient is in sternal position. Someone should sit with the patient while eating. It is beneficial for the patient to maintain an upright position for 30 minutes after feeding to decrease the risk of regurgitation and aspiration pneumonia.
- Patients who are at high risk of, or have radiographic confirmation of aspiration or hypostatic pneumonia should undergo coupage each time the patient is turned if aspiration or hypostatic pneumonia is suspected; however, this is not always well tolerated.



Figure 11.6 Coupage is gentle percussion over the thoracic following the lung field using a cupped hand.



Figure 11.7 Coupage on both sides of chest.

Coupage is gentle percussion over the thoracic cage using a cupped hand (Figures 11.6 and 11.7). It is contraindicated in thoracic trauma patients. Thoracic auscultation should be performed at least once daily to identify abnormalities; these should be reported to the veterinarian immediately. Postural therapy with the patient supported in various positions (e.g., sitting, over a therapy ball rolled between head down–rear elevated and head up–rear down) can also be implemented to aid removal of excess secretions in combination with nebulization and coupage.

Urination

As a rule, recumbent patients should have general bladder care. If they are paretic, they should not have difficulty urinating, but



Figure 11.8 Manual expression of the bladder is a learned skill that the rehabilitation veterinary technician should know and can teach.

sometimes cannot move away from the urine adequately. If they are plegic, they are likely unable to urinate voluntarily and require bladder assessment or management in the form of catheterization or manual expression (Figure 11.8). In neurological patients, urinary function often returns as soon the patient has voluntary motor function, but this is not a predictable pattern (Olby *et al.*, 2003). This primarily applies to patients with thoracolumbar disease (only pelvic limbs affected). Patients with cervical disease (all four limbs affected) may be reluctant to urinate as they are unable to adopt a posture for urination. Diseases of the lumbosacral spinal cord are a slight exception as these patients

Box 11.3 Bladder palpation and expression

If the patient is recumbent, then the owner must be taught how to palpate the bladder, as well as how and when to express it.

- 1) Using one or both hands, depending on the patient's size, place fingers on both sides of the abdomen
- 2) Start just below the ribs and squeeze gently inward
- 3) Slide fingers caudal until the bladder is located; palpate for size and tone (the bladder has been described as a water balloon in the abdomen).
- 4) Apply gentle pressure to express the urine

Avoid expression of the bladder if:

- the bladder is small (this will allow it to fill and the patient is more likely to feel the need to urinate); or
- the bladder has a strong tone (it may have lost its elasticity from being overdistended for too long or may be an upper motor neuron bladder that requires medication to relax the detrusor and urethral sphincter muscles).

Note: If the bladder must be expressed frequently then this could be painful.

Adapted from Weber (2016).

have urinary difficulties despite retaining voluntary motor function.

An understanding of the urinary function is critical to judge whether urination is voluntary or not. Each time the pressure in the bladder exceeds that of the urethral sphincter, urine will leak out and this may be misinterpreted as voluntary urination (Box 11.3). Thus, other measures need to be taken to assess the presence of voluntary urination. It is important to teach those caring for the patient how to palpate the bladder and monitor for appropriate urination.

Assessment of the bladder includes the following:

- Palpation to assess bladder size before and after urination (training in palpation of bladders is important).
- Recording of all urination in the medical record noting whether voluntary, expressed manually or via a catheter.
- Preferably urinalysis on admission and testing urine with a dipstick every 2–4 days for the presence of white blood cells and protein.

Appropriate bladder management in recumbent patients includes the following (Box 11.4):

- Regular visits outside to encourage the patient to urinate, at least 3 times daily.

Box 11.4 Monitoring urine stream and quantity

Owners should be asked to monitor their patient's urination by answering a few basic questions:

- Can the pet posture to urinate?
- Does the stream come out normally (forceful) or does it come out in dribbles or sporadic?
- Is this a normal amount of urine or is it decreased?
- Does the urine have an odor?
- Is the urine a normal color?

Note: Any patient that cannot urinate should be reported to the rehabilitation veterinarian immediately.

Adapted from Weber (2016).

A Help 'em Up Harness™ or similar device can be very helpful for maneuvering the recumbent patient if they are medium to large size.

- If unable to urinate, three bladder management options are available: manual expression every 6–8 hours (depending on bladder size), intermittent catheterization every 6–8 hours (depending on bladder size), or placement of indwelling catheter (urine bag to be checked every 4–6 hours and output recorded).

- Medications to improve bladder function will be prescribed by the veterinarian if appropriate.
- Bedding must allow liquid to be absorbed and pass through away from the patient's skin (e.g., acrylic bedding). If incontinence sheets are used directly under the patient's skin, they must fulfill this criterion too.
- Keep the patient clean and dry always, clip long hair if required to enable hygiene management and to allow accurate assessment of urine scalding developing/progressing.

Defecation

Fecal incontinence can occur because the external anal sphincter has been anatomically disrupted, or the nerves damaged or destroyed. Nerve damage, spinal cord disease, or a degenerative disorder may reduce or stop sphincter function. Fecal incontinence affects dogs with severe lumbosacral disease, leading to a lack of voluntary control over defecation and severe soiling. Cats with neurological problems have a tendency toward constipation. It is important to keep the patient clean and dry always. Lactulose may be used (especially in cats) to aid defecation if constipation is suspected, as manual evacuation is difficult in cats. Patients receiving opioid analgesia should be monitored closely for constipation due to reduced intestinal motility, and pelvic trauma patients should be monitored for tenesmus. In both cases treatment by the rehabilitation veterinarian to aid in defecation may be needed.

Skin Care

Recumbent patients are at risk of development of dermatitis secondary to urine scald and fecal soiling and the development of decubital ulcers over pressure points. In addition, skin lesions can develop if the patient is dragging themselves or a limb over rough ground.

Complications can be prevented by use of appropriate soft bedding that absorbs the

liquid. This includes using incontinence pads; however, care must be taken to avoid placing the pad directly beneath the patient's skin as the urine will only disperse across the pad, resulting in increased contact time and leading to urine scalds. Acrylic absorbent bedding should be placed directly beneath the patient, followed by the incontinence pad. This prevents multiple layers of bedding becoming soiled and avoids the recumbent patient lying in their own urine. Ensure appropriate padding is used around pressure points and perform systematic bony point checks twice daily to monitor for skin redness or early development of decubital ulcers. The patient must be turned regularly (every 4–6 hours) and pressure points massaged to increase local blood flow. If necessary, clip the hair in the perineal region. There should be prompt removal of soiled bedding. See earlier section about appropriate bladder management. The patient must always be kept dry and clean.

Treatment of skin complications includes the following:

- Cleaning of dermatitis with a dilute chlorhexidine solution followed by thorough drying and application of a barrier cream.
- Avoiding excessive moisture around affected areas and the application of thick layers of barrier cream, which will only harbor and insulate bacteria. A thin layer of barrier cream that can be easily wiped off is more suitable.
- Applying a dilute solution of bicarbonate of soda and cooled boiled water. This is very effective as it neutralizes the acidity of urine on the skin so reducing scalds or skin irritation. Carefully make sure it is applied into skin folds (for example those of the testes). The area should be doused and left to dry at room temperature. This can be repeated 3–4 times daily.
- Ensuring that, if decubital ulcers develop, pressure is no longer placed over that region. This can be done by using a doughnut-shaped cushion (in some cases suturing to surrounding skin).

- Debridement of necrotic tissue.
- Elizabethan collars to prevent the patient licking or chewing the region.

Care of Toenails

Toenails should be checked and trimmed regularly for the comfort of the patient. Excessively long toenails can hinder the patient from standing/walking when and if they can accomplish this task.

Muscle and Joint Problems

Neurological problems with paralysis can cause muscle atrophy and contracture, and deterioration of joints and their associated soft tissues. Muscle and joint pain can occur secondary to these changes.

Management of these problems includes:

- Treatment of pain.
- An individualized rehabilitation plan that should be started as soon as possible. Exercises for the client at home can be developed.
- Passive range of motion (PROM) and massage to help maintain joint mobility.
- A nutritional plan.

Attitude or Demeanor

Recumbency can contribute to frustration and anxiousness, which can be heightened by pain (Lorenz *et al.*, 2011b). Demeanor may be improved by the following:

- Ensuring the animal's comfort through pain management and appropriate bedding. This includes the amount of contact with humans because some patients prefer to be quiet while others prefer to see and have regular interactions with humans.
- Arranging regular contact with the client.
- Taking the canine patient outside regularly. This is not appropriate for cats unless they are used to harness/leash walking. A cat can have a period of free exploration in a secure cat ward under supervision.

- Placing the patient in a standing position, using support as necessary. Additionally, this helps with pulmonary, musculoskeletal systems, and skin.
- Regular manual therapies (PROM, massage, etc.).

Positioning

The rehabilitation team must be cognizant of the risks facing the recumbent patient. Skin and pulmonary integrity can be compromised if the patient is not on a proper turning schedule. Positioning is important and should be noted on patient charts (Francis, 2007). The patient should be on either side in lateral recumbency, sternal recumbency, sitting and standing, if possible.

Pain

Neurological patients experience pain. Many patients are in pain caused by their primary disease, surgical intervention, or secondary effects of recumbency and spasticity (Lorenz *et al.*, 2011b). The level of pain should be assessed 2–4 times daily depending on the patient. A pain scoring system is most useful for consistency (see Chapter 3).

Manual therapy, ice, heat, electrical stimulation, and therapeutic ultrasound may be used depending on the severity and phase of recovery (acute versus subacute) (Francis, 2007). Because pain-free animals are relaxed and cooperative, recover faster and more completely, and clients are much happier and more compliant with recommendations when their pet is comfortable (Thomas *et al.*, 2014), pain medications should be used to allow for patient comfort.

Nutrition

A recumbent patient's metabolic needs differ from those of a healthy patient (Weber, 2016). The following formula is used to calculate daily needs:

Resting energy requirements (RER) (kcal/day)
 $= 70 \times \text{body weight (kg)}^{0.75}$

Recumbent patients may have reduced appetite or even be anorexic because of pain or depression. The patient needs to be encouraged to eat, although overfeeding of the obese patient needs to be avoided. Extra weight impacts joints and spine when recumbent and can delay return to ambulation because more strength is needed to move a body of higher mass. Do not forget patients' daily water requirements; water intake is important to reduce the risk for cystitis and urinary tract infections in down dogs with bladder impairment. Daily water intake should be 50 mg/kg body weight. Encourage patients to drink by adding small amounts of low-sodium chicken or beef broth to their water. Syringing water or administering subcutaneous fluids may also be recommended in some situations (Weber, 2016).

Risks Affecting Hospitalized Recumbent or Neurological Patients

The recumbent or neurological patient in hospital is at risk of a range of conditions that the veterinary technician/nurse should be alert to (Abramson, 2009):

- Prolonged or permanent loss of mobility and independence secondary to disuse atrophy
- Chronic pain
- Decubital ulcers
- Urine scald
- Depression
- Self-inflicted trauma
- Reduced lung capacity and compliance
- Obesity.

Key Therapeutic Points

In addition, the following key therapeutic points should be noted (Davidson, 2009):

- Bladder care must be initiated for incontinent animals to prevent atony and treat infections.

- Attention to bedding and hygiene helps prevent decubital ulcers.
- Neuromuscular electrical stimulation may be used to strengthen muscle.
- Massage can reduce muscle spasms and pain.
- PROM can be used to maintain joint motion and health.
- Assisted standing, balancing, and various types of exercise can be incorporated, depending on the animal's neurologic status.

Types of Neurological Conditions

It is important that the veterinary technician or nurse is familiar with neurological conditions that affect their patients. Being able to discuss the disease with the client, so that the rehabilitation veterinarian's protocols may be carried out, is imperative. Neurological conditions can affect the animal's mobility, balance, strength, and coordination as well as contributing to pain or discomfort (Price, 2014a).

Canine Conditions

Intervertebral Disc Disease

Intervertebral disc disease is a syndrome of pain and/or neurological deficits, resulting from herniation or protrusion of the disc. Although a normal disc can herniate because of major trauma, typically herniation is secondary to preexisting degeneration of the disc (Jeffery *et al.*, 2016). Disc degeneration is associated with decreased water and proteoglycan concentration (Steinberg and Coates, 2013). This dehydration reduces the ability of the disc to function as a hydraulic cushion and predisposes to displacement of a portion of the disc.

There are two types of disc degeneration: chondroid and fibrous metaplasia. Chondroid metaplasia is most common in chondrodystrophic breeds and causes a progressive

transformation of the gelatinous nucleus pulposus to hyaline cartilage. This type of degeneration predisposes to disc extrusion where disc material is displaced beyond the outer edge of the annulus (also called Hansen type I). Fibrous metaplasia is an age-related degenerative process that occurs in any breed but is more common in non-chondrodystrophic dogs 7 years of age and older. It is characterized by a fibrous collagenization of the nucleus pulposus and degeneration of the annulus fibrosus. Fibrous metaplasia predisposes to disc protrusion, in which disc material is displaced from the disc space but is contained within an intact annulus (also called Hansen type II). Non-chondrodystrophic breed dogs can also suffer extrusions and less commonly chondrodystrophic breeds suffer protrusions (Thomas *et al.*, 2014).

Trauma

Automobile accidents, gunshot wounds, falls, and dog fight injuries are common causes of traumatic spinal cord injuries. Spinal injuries may consist of vertebral luxation, vertebral fracture, vertebral fracture/luxation, contusions, concussions, and hemorrhage, as well as traumatic disc herniation.

Peripheral Nerve Injury

Common causes of peripheral nerve injury include fractures (e.g., femoral fracture that damages the sciatic nerve), intramuscular injection (usually affecting the sciatic nerve), traumatic brachial plexus nerve root avulsion, and iatrogenic causes including swelling, trauma to nerve, and hemorrhage that may sometimes occur in surgery.

Neoplastic Disease

Spinal tumors are classified by their location as extradural, subarachnoid (intradural–extramedullary), and intramedullary. Extradural neoplasia is most common and includes osteosarcoma, multiple myeloma, fibrosarcoma, chondrosarcoma, hemangiosarcoma, and lymphosarcoma. Extradural lymphosarcoma is the most common spinal tumor in cats and may be secondary to infection with the feline leukemia

virus (Thomas *et al.*, 2014). Subarachnoid tumors include nerve sheath tumors, meningioma, lymphosarcoma, and neuroepithelioma. Intramedullary tumors include glioma, lymphosarcoma, and hemangiosarcoma. Metastatic tumors, including metastatic carcinomas and sarcomas, may occur in any location. Spinal neoplasia is more common in middle-aged or older animals (Thomas *et al.*, 2014).

Degenerative Myelopathy

This is a slowly progressive degenerative disease of the CNS that most severely affects the thoracic spinal cord segments. It is inherited as an autosomal recessive trait and is associated with a mutation in the superoxide dismutase 1 (*SOD1*) gene, like the familial form of amyotrophic lateral sclerosis in human patients (Lou Gehrig's disease) (Awano *et al.*, 2009). Degenerative myelopathy affects older dogs, usually more than 8 years of age. German Shepherds, Rhodesian Ridgebacks, Pembroke and Cardigan Welsh Corgis, Boxers, and Chesapeake Bay Retrievers are most commonly affected, although cases have been documented in several other large-breed dogs and cross breeds (Coates and Wininger, 2010).

Myasthenia Gravis

Myasthenia gravis is a neuromuscular disorder (Khorzad *et al.*, 2011) caused by a reduction in the number of functional nicotinic acetylcholine receptors (AChR) on the postsynaptic membrane of the neuromuscular junction. Myasthenia gravis exists either as a rare congenital disease or more commonly as an acquired autoimmune disease, resulting in a deficiency of nicotinic AChR. Acetylcholine is essential for muscle contraction at the neuromuscular junction. The defect in transmission resulting from AChR loss leads to focal or generalized muscle weakness and exhaustion.

Cervical Spondylomyelopathy

Caudal cervical spondylomyelopathy, or wobbler's syndrome, is a disorder caused by abnormal development of the cervical

vertebrae resulting in compression of the spinal cord. Middle-aged (7–9 years) Doberman Pinschers and young (1–4 years) giant-breed dogs (Great Danes, Mastiffs, Rottweilers, Bernese and Swiss Mountain dogs) are most commonly affected (da Costa, 2010).

Fibrocartilaginous Embolic Myelopathy

Fibrocartilaginous embolic myelopathy (FCEM) is an acute infarction of the spinal cord caused by a vascular embolus of fibrocartilage, probably originating from the intervertebral disc. Adult large- or giant-breed dogs and miniature Schnauzers are most commonly affected. This disease is less common in small dogs, chondrodystrophic dogs, and cats. The onset is sudden and often associated with activity such as running or playing. Any region of the spinal cord may be affected, and the spinal cord segments involved dictate the specific neurological deficits. In severe cases, there is loss of deep pain perception caudal to the lesion (Thomas *et al.*, 2014).

Discospondylitis

Discospondylitis is an infection of the intervertebral disc and adjacent vertebral bodies. It is usually caused by hematogenous spread of organisms from sites elsewhere in the body, such as the urinary tract, skin, or mouth. Penetrating wounds, surgery, or plant material migration can cause direct infection of the disc space or vertebra. In certain geographic regions migrating plant material such as grass awns is a relatively common cause of vertebral osteomyelitis and discospondylitis of the L2–L4 vertebrae (Thomas *et al.*, 2014).

Polyneuropathies

Breed-associated polyneuropathies, such as polyneuropathy of hypothyroidism and laryngeal-paralysis associated polyneuropathy are sometimes seen.

Feline Conditions

Intervertebral Disc Disease

Intervertebral disc disease is encountered much more rarely in cats compared with

dogs, although both Hansen type I (disc extrusion) and type II (disc protrusion) herniations have been observed (Sharp, 2012b). Type II cases are the more common in cats, though still rarely found clinically, probably due to the slowly progressive nature of the condition and the lack of obvious clinical signs in this species.

Fibrocartilaginous Embolism

This is rare in cats and, as in dogs, is associated with variable clinical signs and prognosis. A full physiotherapy assessment is essential to identify the specific functional problems of the individual animal and allow the selection of appropriate physiotherapy interventions to treat those problems.

Brachial Plexus Avulsion

This type of traumatic neuropathy is common in cats, with clinical signs including muscle weakness (commonly called “dropped elbow”), proprioceptive deficits (dragging of the foot), and absence of spinal reflexes and deep pain perception. The prognosis is often poor for these animals, but the provision of physiotherapy can help to reduce edema, maintain circulation to the affected area, prevent contractures, and maintain activity in muscles (Bockstahler and Levine, 2011).

Trauma

Automobile accidents, falls, being kicked or abused or fighting injuries are common (Sharp, 2012b).

Tail Pull Injuries

“Tail pull injury” or sacrococcygeal subluxation is a common neurological condition in cats and can also occur in dogs, albeit rarely (Davies and Walmsley, 2012). It occurs most frequently because of a road traffic accident and results in a varying degree of trauma to the sacral spinal cord segments and cauda equina. The neurological deficits associated with sacrocaudal luxation can include paraparesis, but this is usually transient. The most severe sequelae relate to anal sphincter and bladder dysfunction.

Establishing Goals

Long-term goals define the patient's expected level of performance at the end of the rehabilitation process. Short-term goals are the component skills established at each phase of rehabilitation that will be needed to attain the long-term goals. This plan helps establish the at-home treatment plan that is given to clients (Sturges and Woelz, 2005).

Goals

The goal with neurological patients is to challenge them without causing excess neuromuscular fatigue, pushing them but in a positive way with lots of praise and encouragement. The purpose of the exercises is to stimulate proprioceptive tracts to aid body positional awareness and balance, improve muscular strength, and attempt to pattern a correct gait as nerves recover. Typical objectives in rehabilitating the patient with neurological diseases are to minimize pain, reestablish normal neural pathways, prevent secondary complications, and, if possible, return the animal to independent function. Therapy also aims to counteract the effects of inactivity by improving joint fluid circulation and enhancing circulation to tissues (McCauley and Van Dyke, 2013). It is essential not to worsen neurologic function or pain in patients with spinal instability (Thomas *et al.*, 2014). A program that is painful or stressful for the patient is unlikely to be effective, may cause the patient to become aggressive, and will potentially damage the human–animal bond between client and patient (Sims *et al.*, 2015).

Passive Manual Therapies for Dogs and Cats

Passive exercises should be performed for neurological patients who lack voluntary movement or strength or whose proprioceptive deficits preclude a normal gait (Olby *et al.*, 2005).

Passive Range of Motion

PROM is commonly prescribed for neurologically impaired patients, and is frequently combined with stretching of the periarticular connective tissues and skeletal muscle. The primary benefit of PROM is protection against stiffening or fibrosis of the joint. Other benefits may include prevention of cartilage atrophy, replenishment of the synovial fluid (primary source of nutrients for the cartilage), improved local circulation, and stimulation of sensory and proprioceptive pathways in the synovium and periarticular structures (Sims *et al.*, 2015). Placing each joint through a normal range of motion will help to maintain joint health in patients who have deficits in voluntary movement.

PROM should be performed with the patient lying in lateral recumbency on a well-padded surface. The uppermost limbs should be put through gentle flexion and extension of each joint within the patient's comfort zone. Once each joint has been put through 15–20 cycles, each limb may be put through bicycling movements for another 15–20 repetitions. The patient is then flipped and the exercise repeated on the contralateral limbs. This exercise should be performed 3–4 times per day in a recumbent patient.

Massage

Massage is another passive therapy with multiple benefits. It can help to alleviate pain and stress associated with recent injury or surgery. Massage aims to improve both local and whole-body circulation and lymphatic drainage, allowing for increased tissue oxygenation and more rapid resolution of edema. Sensory stimulation that could encourage nerve firing in the affected tissues is an added benefit. Massage is clearly pleasurable for most rehabilitation patients and beginning a therapy session with a whole-body massage in a calming environment may improve patient compliance, thereby increasing the effectiveness of the therapy program. Massage of a mildly contracted muscle group

may also be beneficial in restoring its function and should be performed 2–3 times per day after pre-warming the region (Olby *et al.*, 2005).

Acupuncture

Acupuncture can be considered a passive therapy because it does not require the active involvement of the patient and relies primarily on direct mechanical stimulation to achieve its effects. The scientific literature is mixed regarding the therapeutic benefits of acupuncture overall, but the evidence may be more encouraging when examining the benefits of acupuncture specifically for neurogenic pain and the facilitation of nerve signaling (Sims *et al.*, 2015). Only the veterinarian certified in veterinary acupuncture may perform this therapy.

Therapeutic Exercise for Dogs and Cats

Therapeutic exercise programs focus on proprioception and balance, muscle strengthening, re-education of normal posture, and gait training. Proprioception and balance work is important for the neurologically impaired patient. Weight shift training is used following an injury or postoperatively, initially training the patient to use the affected limb, and later encouraging appropriate weight distribution during activities of daily living. Strengthening can focus on an isolated muscle, limb, or body region (McCauley and Van Dyke, 2013). Posture re-education addresses the static postures (stand, sit down) as well as the transitions between these postures. Gait retraining addresses the patient that does not properly use one or more limbs or that has developed an abnormal gait behavior or pattern.

The rehabilitation veterinarian will be the one that selects which therapeutic exercises will be used for both short- and long-term goals. The rehabilitation veterinary technician or nurse will carry out these exercises

with the patient and will be responsible for teaching the exercises to be used by the client at home.

Targeted exercise therapy aids in the management or prevention of many of the consequences of immobility, including atrophy of soft and bony tissues, stiffening, or fibrosis, maintaining proper proteoglycan matrix in the articular cartilage and stimulating the synovium to replenish the joint fluid (McCauley and Van Dyke, 2013). Exercise therapy stimulates transmission of nerve signals, reinforces proprioceptive and motor pathways, and aids in the restoration of muscle memory for standing, walking, and other activities that require minimal or no conscious effort in the healthy patient (Sims *et al.*, 2015). Exercises that aid in development or preservation of the core muscle groups along the spine and abdomen are particularly helpful in improving a patient's ability to handle transitional movements (Sharp, 2008). The client is often helpful in designing exercises that can be used at home with their own equipment or household items (Box 11.5).

The neurologically impaired animal may require a cart or sling during active assisted range of motion and other exercises. A lift can be used in clinic (Figure 11.9). Support should be used if necessary and may be in the form of splints or orthoses (see Chapter 10).

Therapeutic Modalities for Dogs and Cats

Thermotherapy

Thermotherapy is the use of superficial heat or cold as a therapeutic modality for the treatment of disease or trauma. It may be applied using many different methods (Dragone *et al.*, 2014). It is discussed in detail in Chapter 14.

Therapeutic Ultrasound

The application of therapeutic ultrasound to soft tissues may help to alleviate pain while improving tissue blood supply and speeding

Box 11.5 Therapeutic exercises for the neurological patient

- *Neuromuscular facilitation*—Visualize and perform a passive running pattern with the dog's front or hind leg. Be aware of stimuli under the pad of the foot when appropriate in the cycle. This helps to ingrain the pattern of movement and stimulate the neural pathways connected with this. It can be an aid in neural reprogramming and regeneration (Edge-Hughes, 2013).
- *Vibration*—The use of a small electrical vibrator can help to stimulate muscle contractions in neurological cases (Edge-Hughes, 2013).
- *Ice massage*—Water can be frozen in a styro-foam or plastic cup and then applied to the affected area directly. To perform the massage, the therapist exposes a portion of the ice surface, holds the cup, and applies the ice surface directly to the patient's skin. The ice surface is moved in a continuous, circular fashion across the treatment area for 5–10 minutes (Hanks *et al.*, 2015).
- *Tapping*—Fast and steady tapping over a muscle belly can lead to muscle contractions as well as stimulus of neural receptors in muscles and tendons.
- *Weight bearing techniques*—These are discussed in Chapter 19.
- *Postural reactions (these are not reflexes) from perturbations*—Hopping, parawalking, wheelbarrowing, directional rolling over an inflatable therapy ball, tilt table.
- *Land treadmill exercise*—Active assisted range of motion is possible with the therapist seated over the treadmill behind the patient.
- *Underwater treadmill or supported swimming*—This is discussed in Chapter 18.
- *Supported standing*—This is discussed in Chapter 10.
- *Rhythmical stabilizations*—With the animal standing squarely, the therapist places gentle pressure over the pelvis or cranial thoracic region and gently “bounces” the animal up and down. The bouncing motion should be relatively rapid, with only enough recovery time to regain the normal standing position. This generates rapid firing of postural muscles in an isometric fashion, which is ideal for weakly ambulatory or significantly ataxic patients (Drum *et al.*, 2015).
- *Tensor bandaging*—Wrap an Ace™ bandage loosely around the thoracic and pelvic limbs. This helps to create awareness and connection between the front and hind end.
- *Joint compressions* (veterinarian or physical therapist only)—Place one hand above the joint and the other hand below the joint. Push the two joints together.
- *Tactile sensory stimuli*—Any sensory stimuli you can provide to the superficial skin receptors can be helpful. Examples include brushing, gently pulling hair, or rubbing the hair in the wrong direction, zigzag petting or tapping, pinching, or picking up the skin, thermal stimulus using ice or a hot pack (with care).
- *Clapping over body*—Use your hand in a cupped position so that you do not slap the patient.
- *Wringing the limb*—Gently move your hands circumferentially around the limb of the patient as though you were wringing out a wet towel, up and down the limb (Edge-Hughes, 2013).
- *Acupressure*—This involves application of pressure at acupuncture points. Laser can be used on acupuncture points.

up healing. Ultrasound may be beneficial for epaxial muscles that are experiencing muscle spasms. Its use is contraindicated over an exposed spinal cord, and continuous mode ultrasound is not recommended in postoperative neurosurgical patients (Olby *et al.*, 2005). This technique is discussed in detail in Chapter 17.

Neuromuscular Electrical Stimulation

The application of neuromuscular electrical stimulation (NMES) in patients with acute spinal cord disease may be beneficial to increase tissue perfusion, decrease pain, and delay the onset of disuse muscle atrophy. In patients



Figure 11.9 Working on a patient in a Hoyer lift.

with LMN disease, stimulation of the affected muscle groups will delay the onset and severity of neurogenic muscle atrophy (Olby *et al.*, 2005). The use of electrical stimulation is preferred for muscle groups that are not already experiencing spasms. It is contraindicated over surgical sites following a laminectomy or pediclectomy until adequate healing has taken place. NMES should be applied to the muscle groups of affected limbs once a day for 15 minutes each until the patient is ambulating with mild-to-moderate ataxia. This technique is discussed in detail in Chapter 16.

Laser Therapy

Laser therapy has been increasingly incorporated into rehabilitation programs for a variety of conditions, including skin wounds; muscle, tendon, and ligament injuries; neurologic conditions; arthritis; and pain (Millis and Saunders, 2014). This modality is discussed in detail in Chapter 15.

A Special Word About Cats

One must think like a cat to have success. The treatment room must be calm. A peaceful environment encourages a relaxed mood in preparation for rehabilitation treatment. Rehabilitation programs can benefit from using a cat's love of play and its natural hunting instincts. This may allow a more hands-off approach, which is beneficial for cats that are particularly aggressive or fearful (Sharp, 2012b).

Keep it Short

All rehabilitation sessions should be kept short to maintain interest and prevent boredom (Price, 2014b). Activities should be introduced at a level appropriate to the recovery of the cat, and progressed to accommodate the cat's improvement.

Make it Fun

Devices that demand activity to variously release food (treat balls), encourage the tracking/hunting instinct (toys, motorized units), or that involve reaching observational positions (climbing towers, window sills) can all be beneficial if used appropriately. Placing or hiding food in positions that encourage activity to access it (top of stairs, top of climbing frame, requiring climbing over/under obstacles and/or through tunnels), or simply moving food regularly between rooms can all increase activity levels (Sharp, 2012b). Play combines elements of predatory behavior, including exploration, investigation, stalking, chasing, attacking, pouncing, leaping, swatting, and grasping, and the cat can be provided with a variety of targets (Figure 11.8). Toys that bounce or move in ways that entice the cat to play can be usefully provided, and successful interactive toys might include zig-zag balls, mice on elastic string, or fishing rod toys with fur or feathers. Moving spots of light produced by mirrors, laser pointers, or torches might stimulate some cats to chase,

although limiting play just to chasing lights can lead to frustration as the chase is never successfully concluded and the cat is never able to catch and “kill” the target.

Create an Enriched Environment

The development of an enriched environment can provide a valuable means of exercising cats in a “hands-off” manner. Creating a small area (or using a small room) filled with obstacles, as well as hidden treats or toys, may encourage an inquisitive cat to explore and find the hidden “gems” (Sharp, 2012a). A variety of obstacles should be included that require the cat to climb over, crawl under, and creep through them to promote joint mobility, balance, and general activity. Scratching posts can encourage joint flexibility and strength (Sharp, 2012b). The inclusion of various surfaces (bean bags, cushions, pillows) can provide sensory input as well as promoting strength and balance as the cat walks over them to explore the area.

Client Education and Support During Home Therapy

The approach is to teach as much as you can to clients so that the treatment of their pet can be carried over into the home. Time is taken to educate clients about their pet’s condition and home exercises and instructions are given so they can help their pet recover faster at home. The rehabilitation veterinary technician or nurse plays a key role in teaching the client how to manage the needs of a patient with limited mobility, possibly more so than any other clinician involved in the medical management of patients with neurologic injury or disease (Sims *et al.*, 2015). Clients bear the greatest responsibility for patient care once discharged from the hospital. They should be informed about the patient’s nursing care needs and signs of complications that need to be addressed by

their primary veterinarian. A thorough description, ideally with visual or written aids, of any home therapies will improve the client’s confidence and increase compliance.

The technician should help the therapist evaluate the client’s ability to understand and safely perform the recommended treatments. Financial, physical, or scheduling restrictions facing the client or pet in the home environment should be discussed because they impact the type, number, and frequency of any prescribed treatments. In cases where the family schedule or home environment cannot accommodate the needs of the patient in the near term, the therapist may recommend hospitalizing or boarding the patient for inpatient care during the initial phases of the physiotherapy program. The use of readily available or inexpensive materials minimizes the burden on the client to seek out or develop equipment for their pet’s therapy.

It is important to include information such as a description or demonstration of the activities to be performed, how frequently the treatments should be performed, signs that indicate a treatment is not well tolerated or ineffective, a basic understanding of relevant anatomy, and an approximate timeline for the anticipated results. The rehabilitation team should be prepared not only to provide guidance on medical and nursing care, but also to address concerns related to patient welfare, related to the emotional, physical, and financial burdens of managing a pet with special needs, and related to quality-of-life and end-of-life decisions.

Many clients are overwhelmed with the degree of their pet’s disability and the tremendous impact of their care on the home routine. Often, clients do not reach a full realization of their new responsibilities until several days or weeks after the diagnosis is obtained and the patient is discharged from the hospital. The therapist needs to set realistic expectations for the client pertaining to recovery or the rate of decline expected. Clients will greatly benefit from referral to a rehabilitation team who can help manage the emotional fallout from caring for a paralyzed or debilitated pet (see also Chapter 12).

Conclusions

Neurological rehabilitation can be among the most challenging and rewarding work for the veterinary rehabilitation team. Determining time for recovery is often the most difficult task. It is important to remember that recovery times can be extremely variable, and are intrinsically linked to the neurologic condition, underlying medical conditions, and neurologic status upon pres-

entation for rehabilitation (Drum, 2010). One must consider time available for treatment, both for the veterinary team and the client, as it is often not feasible to perform all exercises and modalities in a single patient. Some exercises may not be applicable or possible for some patients. Each patient requires a rehabilitation protocol that is specifically designed for that patient's neurologic condition, client expectations and level of participation, and expertise of the veterinary team.

References

- Abramson CJ (2009) Nursing care for the "down" dog. In *Proceedings of the American Animal Hospital Association Conference*, Phoenix, Arizona, pp. 721–722.
- Awano T, Johnson GS, Wade CM, *et al.* (2009) Genome-wide association analysis reveals a SOD1 mutation in canine degenerative myelopathy that resembles amyotrophic lateral sclerosis. *Proc Natl Acad Sci USA* 106(8):2794–2799.
- Bockstahler B and Levine D (2011) Physical therapy and rehabilitation. In *The Feline Patient*, 4th edn. (eds. GD Norsworthy, GS Fooshee, MA Crystal, *et al.*). Wiley/Blackwell, Ames, IA, pp. 687–690.
- Brody LT (1999) Mobility impairment. In *Therapeutic Exercise: Moving Toward Function* (eds. CM Hall and LT Brody). Williams and Wilkins, Philadelphia, PA, pp. 57–83.
- Calvo G (2012) Rehabilitation nursing goals. Presented at the WSAVA/FECAVA/BSAVA World Congress, April 12–15, 2012, Birmingham, UK.
- Coates JR and Wininger FA (2010) Canine degenerative myelopathy. *Vet Clin North Am Small Anim Pract* 40(5):929–950.
- Coleman MP and Freeman MR (2010) Wallerian degeneration, Wld, and Nmnat. *Annu Rev Neurosci* 33(1):245–267.
- Davidson JR (2009) Rehabilitation of spinal cord injury. Presented at the 81st Western Veterinary Conference, February 15–19, 2009, Las Vegas, NV.
- Davies E and Walmsley G (2012) Management of tail pull injuries in cats. *In Practice* 34:27–33.
- da Costa RC (2010) Cervical spondylomyelopathy (wobbler syndrome) in dogs. *Vet Clin North Am Small Anim Pract* 40(5):881–913.
- De Lahunta A and Glass E (2009) The neurologic examination. In *Veterinary Neuroanatomy and Clinical Neurology*, 3rd edn. Saunders/Elsevier, St. Louis, MO, pp. 487–501.
- Dragone L, Heinrichs K, Levine D, *et al.* (2014) Superficial thermal modalities. In *Canine Rehabilitation and Physical Therapy*, 2nd edn. (eds. D Millis and D Levine). Elsevier/Saunders, Philadelphia, PA, p. 312.
- Draper WE, Schubert TA, Clemmons RM, and Miles SA (2012) Low-level laser therapy reduces time to ambulation in dogs after hemilaminectomy: a preliminary study. *J Small Anim Pract* 53(8):465–469.
- Drum MG (2010) Physical rehabilitation of the canine neurologic patient. *Vet Clin Small Anim* 40:181–193.
- Drum MG, Marcellin-Little DJ, and Davis MS (2015) Principles and applications of therapeutic exercises for small animals. *Vet Clin Small Anim* 45:73–90.
- Edge-Hughes L (2013) *Therapeutic Exercises for the Neurological Patient*. The Canine Fitness Centre, Calgary, Alberta, Canada. www.caninefitness.com
- Francis M (2007) Rehabilitation for patients with neurological diseases. *Proceedings*

- from ACVIM Conference, June 6–9, 2007, Seattle, WA.
- Gandini G, Cizinauskas S, Lang J, *et al.* (2003) Fibrocartilaginous embolism in 75 dogs: clinical findings and factors influencing the recovery rate. *J Small Anim Pract* 44(2):76–80.
- Hanks J, Levine D, and Bockstahler B (2015) Physical agent modalities in physical therapy and rehabilitation of small animals. *Vet Clin Small Anim* 45:29–44.
- Jeffery ND, Barker AK, Hu HZ, *et al.* (2016) Factors associated with recovery from paraplegia in dogs with loss of pain perception in the pelvic limbs following intervertebral disk herniation. *J Am Vet Med Assoc* 248(4):386–394.
- Joaquim JG, Luna SP, Brondani JT, *et al.* (2010) Comparison of decompressive surgery, electroacupuncture, and decompressive surgery followed by electroacupuncture for the treatment of dogs with intervertebral disk disease with long-standing severe neurologic deficits. *J Am Vet Med Assoc* 236(11):1225–1229.
- Jorge LL, Mota do Nascimento de Brito A, and Garcia Marchi FH (2014) New rehabilitation models for neurologic inpatients in Brazil. *Disabil Rehabil* 37(3):268–273.
- Kathmann I, Cizinauskas S, Doherr MG, *et al.* (2006) Daily controlled physiotherapy increases survival time in dogs with suspected degenerative myelopathy. *J Vet Intern Med* 20(4):927–932.
- Khorzad R, Whelan M, Allen Sisson A, and Shelton GD (2011) Myasthenia gravis in dogs with an emphasis on treatment and critical care management. *J Vet Emerg Crit Care* 21(3):193–208.
- Lorenz MD, Coates JR, and Kent M (2011a) Neurologic history. In *Neuroanatomy, and Neurologic Examination*. Elsevier/Saunders, St. Louis, MO, p. 2.
- Lorenz MD, Coates JR, and Kent M (2011b) Pain. In *Handbook of Veterinary Neurology*, 5th edn. Elsevier/Saunders, St. Louis, MO, p. 429.
- McCauley LM and Van Dyke JB (2013) Therapeutic exercises. In *Canine Sports Medicine and Rehabilitation* (eds. MC Zink and JB Van Dyke). John Wiley & Sons, Inc., Ames, IA, p. 152.
- Millis DL and Mankin J (2014) Orthopedic and neurologic evaluation. In *Canine Rehabilitation and Physical Therapy*, 2nd edn. (eds. D Millis and D Levine). Elsevier/Saunders, Philadelphia, PA, p. 193.
- Millis D and Saunders DG (2014) Laser therapy in canine rehabilitation. In *Canine Rehabilitation and Physical Therapy*, 2nd edn. (eds. D Millis and D Levine). Elsevier/Saunders, Philadelphia, PA, p. 359.
- Olby N (2010) Patients with neurological disorders. In *BSAVA Manual of Canine and Feline Rehabilitation, Supportive and Palliative Care* (eds. S Lindley and P Watson). BSAVA Publications, Gloucester, UK, p. 169.
- Olby N, Levine J, Harris T, *et al.* (2003) Long-term functional outcome of dogs with severe injuries of the thoracolumbar spinal cord: 87 cases (1996–2001). *J Am Vet Med Assoc* 222(6):762–769.
- Olby N, Halling KB, and Glick TR (2005) Rehabilitation for the neurologic patient. *Vet Clin Small Anim* 35:1389–1409.
- Olby NJ, Lim JH, Babb K, *et al.* (2014) Gait scoring in dogs with thoracolumbar spinal cord injuries when walking on a treadmill. *BMC Vet Res* 10:58.
- Panko J (2014) In-hospital rehabilitation of recumbent and neurologic patients. *NAVTA J* June/July:18–25.
- Price H (2014a) Feline physiotherapy: Part 1. *Companion Anim* 19(7):374–378.
- Price H (2014b) Feline physiotherapy: Part 2. *Companion Anim* 19(9):474–478.
- Rylander H (2013) The neurologic examination in companion animals Part 1: Performing the examination. *Today's Vet Pract* January/February:18–22.
- Sharp B (2008) Physiotherapy in small animal practice. *In Pract* 30:190–199.
- Sharp B (2012a) Feline physiotherapy and rehabilitation: 1. Principles and potential. *J Feline Med Surg* 14:622–632.
- Sharp B (2012b) Feline physiotherapy and rehabilitation: 2. Clinical application. *J Feline Med Surg* 14:633–645.

- Sims C, Waldron R, and Marcellin-Little DJ (2015) Rehabilitation and physical therapy for the neurologic veterinary patient. *Vet Clin Small Anim* 45:123–143.
- Steinberg HS and Coates JR (2013) Diagnosis of and treatment options for disorders of the spine. In *Canine Sports Medicine and Rehabilitation* (eds. MC Zink and JB Van Dyke). John Wiley & Sons, Inc., Ames, IA, pp. 311–337.
- Sturges BK and Woelz J (2005) Physical rehabilitation for the neurological patient. Presented at Veterinary Neurology Symposium, University of California Davis, Irvine, CA.
- Thomas WB, Olby N, and Sharon L (2014) Neurologic conditions and physical rehabilitation of the neurologic patient. In *Canine Rehabilitation and Physical Therapy*, 2nd edn. (eds. D Millis and D Levine). Elsevier/Saunders, Philadelphia, PA, p. 609.
- Thomson C and Hahn C (2012) Regional anatomy. In *Veterinary Neuroanatomy: A Clinical Approach*. Saunders/Elsevier, Edinburgh, p. 12.
- Weber DM (2016) Care tips for down dogs. *Veterinary Team Brief* March: 31–35.

12

The Disabled Patient Part 3: Special Considerations for Geriatrics

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Introduction

Gerontology is the study of aging. As veterinary patients age, they experience senescence (the condition or process of deterioration with age), characterized by decreased function and degeneration (degradation) of their bodies. These changes cause a slow decline and can lead to functional issues such as reduced fine motor control and weakness. The aging

process is not a disease process but a decline in reserves and strength as well as other changes in all body systems which increases the likelihood of a patient developing disease. Medical issues commonly encountered in geriatric patients include obesity, degenerative joint disease, neoplasia, and endocrine disease.

A comprehensive medical evaluation is an important requirement in planning rehabilitation for geriatric patients regarding expected

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progression of their physical limitations and how they will respond to physical rehabilitation. In addition to a thorough examination for concurrent disease, nutritional assessment and evaluation of the patient's pain level is important in guiding therapy. Clients with geriatric animals will likely be concerned with quality of life issues, such as mobility, pain level, fecal and urinary continence, appetite, and maintenance of human–pet interaction within the household. These issues can be further complicated by behavioral changes or cognitive dysfunction (Hedges, 2014).

Physical rehabilitation for the geriatric patient entails addressing issues such as pain with care, to avoid exacerbation of weakness and any cognitive signs. Rehabilitation aims to improve mobility and function (via assistive devices or treatments targeted at improving muscle strength and coordination), and address medical and nursing care issues (such as providing adequate bedding and maintaining patient cleanliness in the household). Creating a geriatric wellness program may be beneficial in the physical rehabilitation environment. Marketing aimed at geriatric patients can be targeted to clients interested in maintaining quality of life for their elderly pets, patients with degenerative joint disease, patients with neurogenic issues, and those in need of hospice care.

The Aging Process

Aging is a term that refers to a complex set of biological changes that result in a progressive reduction of the ability to maintain homeostasis when exposed to internal physiologic and external environmental stresses (Goldston, 1995; Bellows *et al.*, 2015). These changes ultimately lead to decreased vitality, increased vulnerability to disease, and eventually death. The landmark study in the *Journal of the American Veterinary Medical Association* looked at large- and medium-breed dogs and identified normal age-related changes (Bellows *et al.*, 2015).

When Do Dogs and Cats Qualify as Senior or Geriatric?

Veterinary professionals consider dogs to be senior at an earlier age than pet owners. Small, medium, and large dogs were all considered senior at around 7 years old by veterinary professionals, with giant-breed dogs hitting the senior mark around 5. Veterinary professionals considered small and medium dogs to be geriatric around 11, large dogs near 9, and giant breeds around 7 (Seymour, 2014) (Table 12.1). The highest percentage of veterinary professionals considered cats to be senior around 9 years old and geriatric around 13 (Seymour, 2014) (Table 12.2). More than 77% of veterinary professionals and pet owners alike said there was a difference between the terms “senior” and “geriatric” (Seymour, 2014).

Physiologic Changes in Aging Dogs and Cats

Aging affects the entire body and its functions (Bellows *et al.*, 2015). There is a loss of lung elasticity and lung capacity is decreased in older pets. The brainstem control of breathing also changes with age and this results in reduced ability to respond to increased demand; older patients become less tolerant of exercise. Panting tends to become less efficient with age. It is harder for older dogs to cool themselves, and this needs to be considered when exercising an elderly dog. Cardiac output starts to decline in mid-life in dogs and again ability to respond to the stresses of exercise may be reduced. Chronic heart valve disease is found in around 30% of older dogs (Box 12.1).

Changes to the integument include changes to skin and nails, such as callous formation, resulting in painful weight bearing and altered foot placement (Figure 12.1), and a tendency to form lumps and bumps such as lipomas, adenomas, etc. Many organ systems are known to decline with age. Changes

Table 12.1 How old is my dog in human years?

Dog's age	0–20 lb (0–9 kg)	21–50 lb (9–23 kg)	51–89 lb (23–40 kg)	>90 lb (>41 kg)
5	36	37	40	42
6	40	42	45	49
7	44	47	50	56
8	48	51	55	64
9	52	56	61	71
10	56	60	66*	78*
11	60	65	72*	86*
12	64	69*	77*	93*
13	68	74*	82*	101*
14	72*	78*	88*	109*
15	76*	83*	93*	115*
16	80*	87*	99*	123*
17	84*	92*	104*	
18	88*	96*	109*	
19	92*	101*	115*	

* = geriatric.

Source: Adapted from Tomlinson (2016).

occur in the gastrointestinal system (reduced elasticity of the colon and rectum), and drug metabolism and elimination by the liver and kidneys becomes slower.

Changes in the immune system result in reduced ability to combat infection and healing can take longer. Geriatric patients are more prone to developing chronic urinary tract infections, for example. Metabolic systems are more likely to be affected with age, for example the adrenal glands undergo changes resulting in altered stress response via increased cortisol production. There may also be changes in electrolyte balance. Other endocrine changes shown by many aging dogs include a dip in thyroid levels. There is also an age-related decrease in basal metabolic rate so obesity risk is higher (Boxes 12.2 and 12.3).

As the nervous system and associated special senses age, changes are commonly seen. Some age-related diminished hearing may occur, although this is difficult to assess in dogs and cats as equipment to test hearing is

not widespread. There can be age-related retinal degeneration. Night vision loss is very common in older dogs and close questioning of the client can establish this. Home adaptations using lighting can help the pet remain active and healthy.

Brain size reduces with age due to atrophy of the cerebral cortex. This is the part of the brain that controls the body's voluntary actions, so patients may be slower to react and slower to learn. There is a loss of fine motor control and decreased body awareness (proprioception). This may result in difficulty with balance and coordination, along with changes in gait. Aged dogs can have slower movements; however, ataxia is never a normal age-related change (Box 12.4).

Mobility and strength decline naturally with age. Sarcopenia is the term for loss of muscle mass due to degeneration with age and this has been confirmed in a study of Labrador Retrievers (Hutchinson *et al.*, 2012). The number and size of muscle fibers decreases and it takes muscles longer to

Table 12.2 How old is my cat in human years?

Life stage	Age of cat	Human equivalent
Kitten (birth to 6 months)	0–1 month	0–1 year
	2–3 months	2–4 years
	4 months	6–8 years
	6 months	10 years
Junior (7 months to 2 years)	7 months	12 years
	12 months	15 years
	18 months	21 years
	2 years	24 years
Adult (3–6 years)	3 years	28 years
	4 years	32 years
	5 years	36 years
	6 years	40 years
Mature (7–10 years)	7 years	44 years
	8 years	48 years
	9 years	52 years
	10 years	56 years
Senior (11–14 years)	11 years	60 years
	12 years	64 years
	13 years	68 years
	14 years	72 years
Geriatric (15 years+)	15 years	76 years
	16 years	80 years
	17 years	84 years
	18 years	88 years
	19 years	92 years
	20 years	96 years
	21 years	100 years
	22 years	104 years
	23 years	108 years
	24 years	112 years
	25 years	116 years

Source: Adapted from International Cat Care (2016).

Box 12.1 Cardiopulmonary changes in geriatric patients

- Declining cardiac output
- Decreased elasticity of pulmonary tissue
- Increased fibrosis of pulmonary tissue
- Decreased cough reflex
 - Pulmonary secretions have low viscosity
 - Increased chance of respiratory disease
- Monitor for pneumonia
- Pulmonary fibrosis in West Highland White Terriers
- Chronic bronchitis in older small breed dogs
- Congestive heart failure exercise intolerance and slowly progressive cough
- Atrioventricular valve thickening in 75% of dogs over 16 years old
- Laryngeal paralysis – can be associated with polyneuropathy, neoplasia, and hypothyroidism
- 32 dogs with idiopathic laryngeal paralysis – 100% had esophageal dysfunction
- 1 year later of the 24 dogs evaluated 100% had neurologic signs (ataxia, weakness, CP (conscious proprioceptive) deficits, muscular atrophy)
- Therefore, start exercise at diagnosis.

Source: Adapted from McCauley (2016a).

Figure 12.1 Callus formation can result in painful weight bearing. A coarse file or sanding tool used lightly can remove callus build-up for more comfortable weight bearing.



Box 12.2 Metabolic changes in the geriatric patient

When creating programs, it is important to take into account:

- Decreased metabolic rate
- Maintenance energy requirement decrease by approximately 20%
- Senior diets typically have reduced calories, but need to provide increased highly digestible protein

Source: Adapted from McCauley (2016a).

respond to exercise. Old dogs frequently adopt a movement pattern of either moving or laying down; they rarely stand in one place. The first muscles to atrophy are the slow twitch postural muscles (e.g., spinal epaxial muscles) as these are very sensitive to reduced load. With atrophy comes decreased ability to maintain a posture and so the process becomes cyclical (Boxes 12.5 and 12.6).

Cognitive Impairment and Other Behavioral Issues

Cognitive dysfunction is a neurodegenerative disorder of senior dogs and cats that is characterized by gradual cognitive decline over a prolonged period (18–24 months or longer) (Landsberg and Araujo, 2005). Diagnosis of cognitive dysfunction syndrome

Box 12.3 Immune changes in the geriatric patient

- Immune compromise
- Decreased bone marrow proficiency
- Diminished phagocytic ability of neutrophils
- Fewer white blood cells and peripheral lymphocytes
- Lymphocytic proliferation responses to stimulation decline
- Decreased CD4:CD8 (helper T:cytotoxic T cell) ratio
- Decreased cellular immunity
- Slower wound healing
- Changes in hair coat
- Aged dogs maintain humoral immune response
 - Respond to vaccines
 - Respond to novel antigenic stimulation

Source: Adapted from McCauley (2016a).

(CDS) is based on recognition of behavioral signs and exclusion of other medical conditions and drug side-effects, which in some cases can mimic or complicate CDS (Warnes, 2015a). Clinical categories include disorientation, alterations in social interactions, sleep–wake cycles, elimination habits, and activity, as well as increasing anxiety. Deficits in learning and memory have also been well documented (Landsberg *et al.*, 2012) (Box 12.7). The diagnosis of CDS was initially

Box 12.4 Nervous system changes in the geriatric patient

- Coordination and proprioception decline
- If muscle strength declines then ataxia may occur (increase falls in humans)
- Loss of vision and vestibular input
- Changes in muscle spindle cells, which provide input on joint position and muscle length for coordinated movement
 - Muscle spindle cells become less sensitive
 - Large myelinated nerve fibers that carry the message from the muscle spindle cell to the spinal cord decline
- Nerve conduction velocity slows (advanced age)
- Tactile sensitivity diminishes in the distal extremities
- Loss of vision due to
 - Cataracts or luxation of the lens
 - Corneal issues, such as keratoconjunctivitis sicca
 - Trauma
 - Retinal issues (degeneration or detachment)
- Loss of hearing.

Source: Adapted from McCauley (2016a).

Box 12.5 Musculoskeletal changes in the geriatric patient

- Sarcopenia—Muscle fibers are replaced with first fat (no loss of muscle circumference), then with fibrous tissue (physical loss of muscle circumference). This causes increased stiffness.
- Reduced oxygenation of muscle fibers
- Loss of strength of muscles and tendons
- Sarcopenia causes
 - Decreased support of the joints
 - Decreased chondrocytes
 - Reduced ability to respond to growth factors
- Can be slowed down by concentric but not eccentric exercises
- Underwater treadmill, balance board, limb weighting, isometric exercises
- Dog study
 - With cage rest see significant atrophy of tibia especially at the insertion of the collateral ligaments
 - Aging decreases tensile strength of ligaments secondary to loss of collagen
- Fun bone facts
 - Animals fully change bone content every 5 years
- Lifetime of weight bearing exercises decrease bone loss normally seen with aging
- Weight bearing training before a bone fracture has faster healing times
- Shockwave therapy may aid bone healing specifically in geriatric dogs
- Water content in the cartilage decreases
 - Thinning of cartilage layer
 - Chondrocytes synthesize smaller, less uniform aggrecan molecules and less functional link proteins
 - Decreased mitotic activities
 - Decrease response to anabolic mechanical stimulants and growth factors
 - Combined with stress on joints secondary to loss of muscle strength leads to eburnation of subchondral bone and osteoarthritis
- Don't see osteoporosis, but bones become more brittle due to infiltration of fat into bone marrow and thinner cortexes
 - Fracture healing is slower
 - More difficult to form a callus
 - Obesity and brittle bones exacerbate progression of osteoarthritis

Source: Adapted from Tomlinson (2016) and McCauley (2016a).

Box 12.6 Muscle changes in the geriatric dog

- Lactic acid builds faster
- Glycogen depletes faster
- Decreased creatine phosphate (used first in muscle use)
- Cannot build muscle strength or endurance to compare to younger dogs
- Muscle capillarization decreases
- Decreased oxygen to muscles
- Decreased endurance
- Type II muscle fibers decrease by ~25%, leading to reduced strength
- Type I muscle fibers can increase or stay the same. These are endurance and postural muscles
- Disuse atrophy—Decreased type I muscle, especially those that cross one joint or postural muscles.

Source: Adapted from McCauley (2016a).

based on clinical signs represented by the acronym DISH, representing Disorientation, altered Interactions with people or other pets, altered Sleep–wake cycles, and House soiling (Landsberg *et al.*, 2013a).

Signs of Cognitive Impairment

Common signs of cognitive impairment include:

- confusion,
- altered relationships and social interactions,
- altered response to stimuli,
- changes in activity: increased anxiety, pacing, repetitive behaviors (vocalizing, pacing),
- changes in activity: apathy, depression,
- altered sleep–wake cycles; reversed day/night schedule,
- learning and memory problems: house soiling,

Box 12.7 Cognitive dysfunction changes in the geriatric patient

- Cognitive changes in the brain—increased oxidative stress
- Deposition of B-amyloid plaques (like in Alzheimer’s patients in humans)
- DNA fragmentation or damage
- Changes in intracellular signaling leading to a loss of neurotrophic factors
- Anatomic changes in the brain
 - Cortical atrophy
 - Increased ventricular volume
 - Reduced neurogenesis in hippocampus (responsible for learning and memory)
- Progressive neurodegenerative disorder
- Diagnosis of exclusion
- Clinical signs include
 - Change in sleeping habits
 - Lack of environmental recognition
 - Decreased interaction with human and animal family
 - Restlessness
 - Apathy
 - Anxiety
- Altered appetite
- Aggression or irritability
- Vocalization
- Incontinence
- Therapy includes
 - Antioxidants—Vitamin B, C, and E; fruits and vegetables
 - Fatty acid supplementation
 - Mitochondrial co-factor supplementation – carnitine, alpha lipoic acid (omega 6 fatty acid), coenzyme Q10
 - Phosphatidylserine (phospholipid that improves cognitive deficits and memory)
 - Ginkgo biloba—a monoamine oxidase inhibitor (MAO-A and MAO-B)
 - Increases dopamine levels and protect neurons from apoptosis induced by B-amyloid
 - Selegiline—an MAO-B inhibitor (Anipryl, L-deprenyl).

Source: Adapted from McCauley (2016a).